Tech Web Hand Book

Basics of Si Power Devices

Powered by ROHM SEMICONDUCTOR

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Introduction

We will be adding new articles related to “power devices” to Tech Web’s “Basic Knowledge”. Recently, discrete devices such as diodes and transistors for handling large power with small loss, known as “power devices” or “power semiconductors”, have gathered attention. The reason is that power devices with higher efficiency and higher performance are needed to address “energy savings” and “miniaturization”, which are issues throughout the world.

However, what specific definitions should we use to categorize these “power devices” that one hears so much about today? Certainly, there are no clear categories, but for example there are diodes and MOSFETs for AC-DC conversion or power switching with high voltage and high power, and then there are power modules that are modularized for output stages of power supply.

When it comes to power devices, new materials such as SiC (silicon carbide) have attracted much attention, but the performance improvement of conventional Si (silicon) power devices, which currently cover a wide range of markets and applications, is also remarkable. This handbook provides basic information on Si diodes and Si transistors, which are classified as power devices.

Since there are so many types of diodes and MOSFETs using Si semiconductors, with a wide range of withstand voltage and current variations, this handbook mainly focuses on the Schottky barrier diode (SBD) and fast recovery diode (FRD) for Si diodes, with power supply applications in mind. For Si transistors, we will focus on their classification and characteristics, as well as high-voltage super junction MOSFETs (SJ MOSFETs).

In addition, we will explain the procedures and decision-making process for selecting transistors for circuit requirements and introduce application examples that utilize the characteristics and features of Si diodes and Si transistors.

1. Si Diodes

This lecture explains the basics of Si diodes through classification, electrical characteristics, and comparison of various types of diodes.

Figure 1. Package image used for power devices

1.1 Types of Si Diodes

In classifying Si diodes, the division is considered to be different depending on the purpose of the classification. As our general framework, we have categorized diodes as rectifier diodes, Zener diodes, and radio frequency diodes. Among these, diodes used mainly for rectification, mainly used for power conversion are further divided into those for general-use rectification, devices for high-speed rectification assuming switching, fast-recovery type diodes for applications in ultra-high speed rectification, and finally Schottky barrier diodes which feature fast operation and low VF. Hereafter each of these is explained in turn.

Although classified in this way, they are all, in principle, elements consisting of an anode and a cathode, and their essential functions and characteristic items (not values) are almost the same. If we are asked "what is the difference?" then I think we can answer "some of the electrical characteristics are optimized for the application.

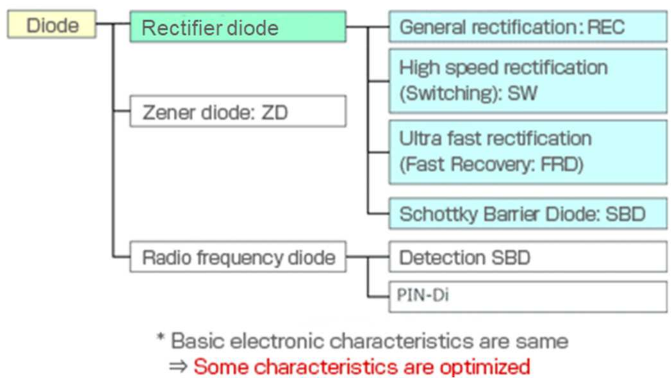


Figure 2. Example of types of Si Diodes

1.2 Electrical Characteristics of Diodes

1.2.1 Static Characteristics

The static characteristics of diodes are shown in Figure 3. The static characteristics of diodes are based on forward voltage VF and forward current IF, and reverse voltage VR and reverse current IR. The orange dashed area in Figure 3 shows the area used by rectifier diodes. Specifically, the IF range that can be handled in the forward direction and the reverse direction will be the area where breakdowns do not occur. It should be noted that the area enclosed by the green dashed line is the usable area of Zener diodes, although these are not discussed in this handbook. This area isn’t usable for other diodes, and if this area is entered without any limits on the IR, device failure may occur.

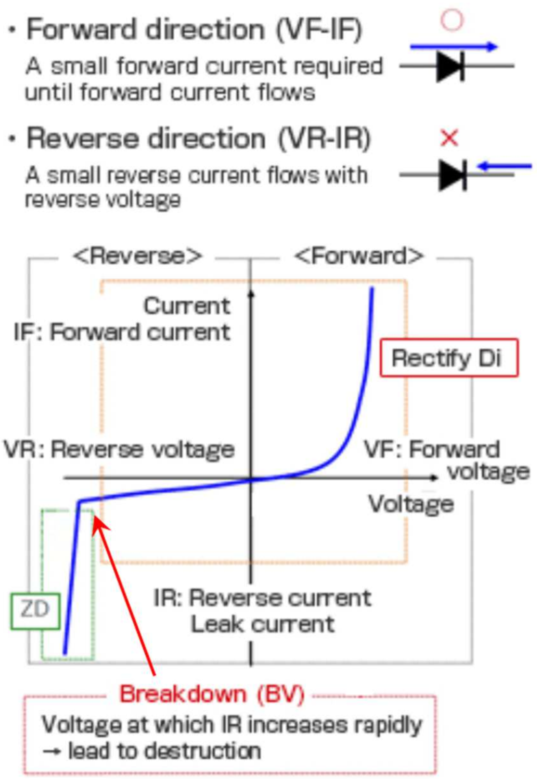


Figure 3. Static characteristics of diodes

1.2.2 Dynamic Characteristics

The main dynamic characteristics of diodes are the reverse recovery time trr, and the static capacitance Ct.

trr is the time when, from the state in which a voltage is applied in the forward direction and forward current IF is flowing, the voltage is changed to the reverse direction and the reverse current IR returns to the steady state (essentially zero). As indicated in the diagram on the right, when the device changes from the ON-state in which an IF is flowing to the OFF-state, ideally IF would immediately go to zero. But in actuality, zero is passed, and a reverse current IR flows momentarily, which recovers to zero in time trr. The shorter trr is, the better is the device characteristic.

The capacitance Ct is the capacitance of the diode itself, and has the same effect as a capacitor. As in the diagram on Figure 4, bottom row, when a diode is turned on and off, if Ct is large, the so-called rounding of the waveform becomes more pronounced, and in some cases there may be the problem that the device begins turn-off operation before an applied voltage reaches a full level due to time constants. In a high-speed switching circuit, diodes with a low Ct are desirable.

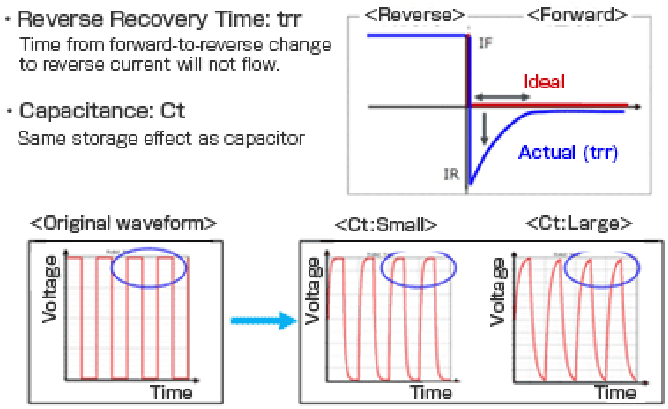


Figure 4. Dynamic characteristics of diodes

1.2.3 Diode Main Characteristics and Maximum Ratings

Each of the above diode characteristics and their major maximum ratings are summarized below. Maximum ratings indicate continuous and instantaneous applied voltages and currents, and the temperature conditions.

For the definition of "instantaneous", either the description on the data sheet should be consulted, or if there is no clear description, the manufacturer should be consulted. At a minimum, you should understand these parameters, as you will be examining them in the actual selection and use of the diode.

● Features Electrical Characteristics of Diodes

VF: Forward voltage

IR: Reverse current (leakage current)

Ct: Capacitance

trr: Reverse recovery time

● Maximum Ratings: Conditions must not be exceeded even momentarily

・Continuous application
Io: Average rectification current
VR: DC reverse voltage

・Instantaneous application only
IFSM: surge current
VRM: Peak reverse voltage

・Operating temperature range
Tj: Junction temperature (internal chip temperature)

・Storage environment of product
Tstg: Storage temperature range

1.2.4 Comparison of the Features of Rectifier Diodes

As shown in Figure 2 in "1.1 Types of Diodes," rectifier diodes can be classified into four categories: general-purpose rectifying applications, general-purpose switching applications, Schottky barrier diodes, and fast-recovery diodes. Table 1 summarizes the characteristics of each of these four types of rectifier diodes.

Type	Features	VF	IR	trr	Price	Suitable applications
Rectification	General purpose	×	○	×	○	General rectification Reverse-connection protection of power supplies
Switching	For switching	×	○	△	△	Simple switching applications Microcomputer peripheral switching
Schottky barrier, SBD	Fast (to 200V) Low VF	○	×	○	×	DC-DC converters AC-DC converters (secondary side)
Fast recovery, FRD	Fast (from 200V)	×	○	○	×	AC-DC converters Inverter circuits

Table 1. Types and Features of Rectifier Diodes

General-purpose rectifier types have as their primary purpose general rectification, that is, conversion of an alternating current into direct current. A diode bridge combines diodes to rectify a current. In addition, diodes are also used for protection so that overcurrents do not flow in the event of reverse-connection of a power supply or battery. The forward voltage VF differs depending on the current that can be handled, but is typically around 1 V. This is the general VF for diodes formed from a silicon PN junction. Rectification of a 50/60 Hz commercial power supply is generally assumed, and so the reverse recovery time trr basically does not need to be fast.

The main application of general-purpose switching-type diodes is, as the name implies, for switching of power supplies. Such devices typically have a VF equal to those of general-purpose rectifier diodes. Because the main purpose is switching applications, trr is shorter than that of general-purpose diodes. However, these devices do not have the speed of Schottky barrier diodes or fast-recovery diodes, but their characteristics are fast compared with general-purpose rectifier diodes.

Schottky barrier diodes (SBD hereafter) do not have PN junctions; instead, they use Schottky barriers, which occur at the junction between a metal and a semiconductor such as N-type silicon. Compared with PN junction diodes they feature a low VF and fast switching characteristics. However, the reverse leakage current IR is large, and depending on conditions, care must be taken to prevent thermal runaway. Even when a large current of for example 10 A is passed, the VF is about 0.8 V, and at several amperes it is around 0.5 V, and so a representative application of these devices is in DC/DC converters and the secondary side of AC/DC converters, where high efficiency is demanded.

Fast-recovery diodes (FRD hereafter) are PN junction diodes, but are fast diodes with a greatly improved trr. And while the rated voltage (reverse voltage VR) of SBDs is usually about 200 V, FRDs can have high voltages of 800 V or higher. However, in general the VF is higher than for general-purpose rectifier diodes, and is typically around 2 V for a device with high voltage, large-current specifications. In recent years however there has been an increase in devices with reduced VFs. Due to their high voltages and fast operation, they are often employed in AC/DC converters and inverter circuits.

The above is illustrated in Figures 5 and 6. Although not something that is limited to these four diode types, the basic temperature characteristics of Si diodes are also shown in Figure 7.

Figure 5 indicates that SBDs have a lower VF and higher IR than the other three types.

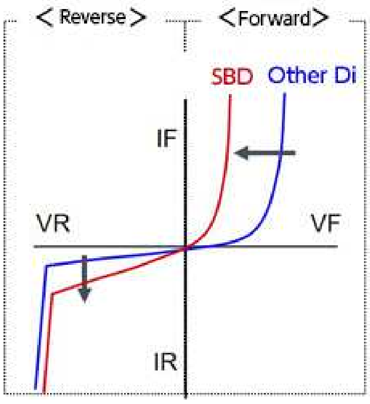


Figure 5. Comparison of current-voltage characteristics between SBDs and other diodes

Figure 6 indicates that trr is much shorter for SBDs and FRDs than for the other two diode types, and that if the IR flowing during the time trr is large, losses are large.

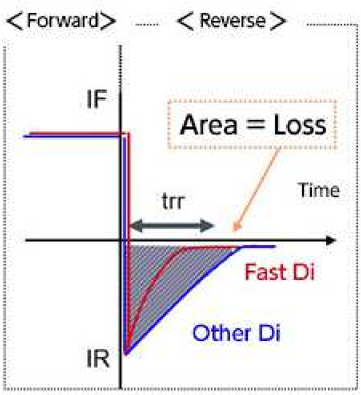


Figure 6. Comparison of trr characteristics between SBDs and FRDs and other diodes

Figure 7 shows the basic temperature characteristics of Si diodes, showing that VF decreases and IR increases at higher temperatures.

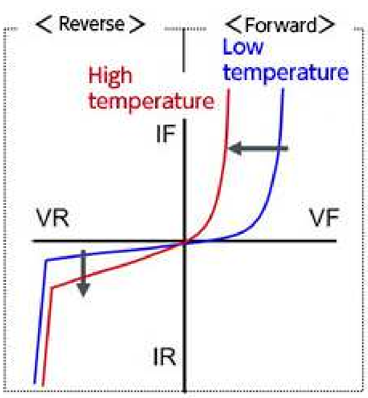


Figure 7. Basic temperature characteristics of Si diodes

1.3 Si Schottky Barrier Diodes (SBD)

This section describes the features and applications of Si SBDs .

1.3.1 Features of Si SBDs

It was explained in section 1.2 that Si SBDs are diodes that use a Schottky barrier resulting from a junction (called a Schottky junction) of the silicon with a metal called a barrier metal, instead of a PN junction. The characteristics of a Si SBD differ depending on the barrier metal type. And, devices may or may not be suited to a given application depending on these differences in characteristics. The following table summarizes the features and suitable applications for different barrier metals. In the table, an “×” indicates that performance is inferior or the device is unsuitable, compared with other devices.

	Metal	VF	IR	trr	High Temp.	Suitable application
Ultra low VF	A	◎	×	○	×	Battery control Undershoot Measures
Low VF & IR (balanced)	B	○	△	○	△	DC-DC Converter
Ultra low IR	C	△	○	○	○	High temp. (eg. automotive)
Other diodes	Non	×	◎	×	◎	General purpose

* SBD characteristics differ depending on barrier metal type

Table 2. Si SBDs: Characteristics of each barrier metal and suitable applications

An SBD that uses barrier metal A features an extremely low VF, but the reverse leakage current IR is somewhat high compared with other devices. Consequently, more heat is generated, making these devices unsuitable when the ambient temperature is high. As explained later, these devices tend to undergo thermal runaway. Because of the low VF, there are low conduction losses and only a small voltage drop, and so these devices are suitable for use in battery-powered circuits.

Barrier metal B-based SBDs balance the VF and IR values. They are often used in DC-DC converter circuits.

SBDs using barrier metal C have extremely low IR values and generate little heat, and so are suited to uses at higher temperatures. For this reason, they are favored in automotive equipment applications.

Other diodes mean general-use PN-junction rectifier diodes. These have small IR values that can be neglected in almost all cases. In contrast, the IR value of some Si SBDs is at a level that requires consideration. This is a point to keep in mind when using Si SBDs.

The following figure shows a graphical representation of the characteristics of each Si SBD represented by ○ and △ in Table 2. For example, for Metal A, the graph of IF-VF characteristics (Fig. 8) shows that VF is very low but IR is large, the graph of IR-VR (Fig. 9) shows that IR is large, but the fluctuation trend where IR increases as VR increases is similar to others, and the graph of IR-VF (Fig. 10) shows that IR is large but VF is small compared to others. The graph of IR-VF (Figure 10) shows that the larger the IR is, the smaller the VF is.

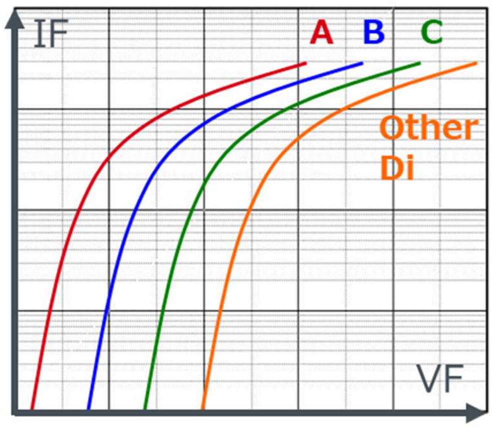


Figure 8. Differences in IF-VF characteristics due to different barrier metals

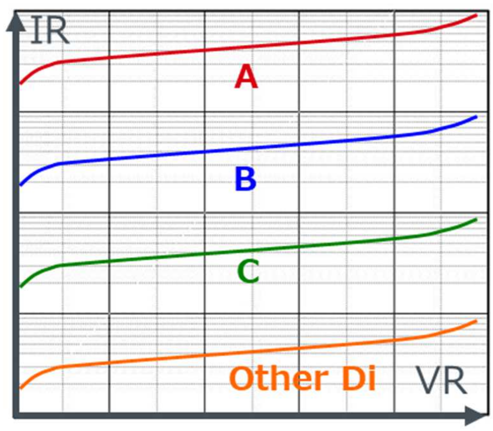


Figure 9. Differences in IR-VR characteristics due to different barrier metals

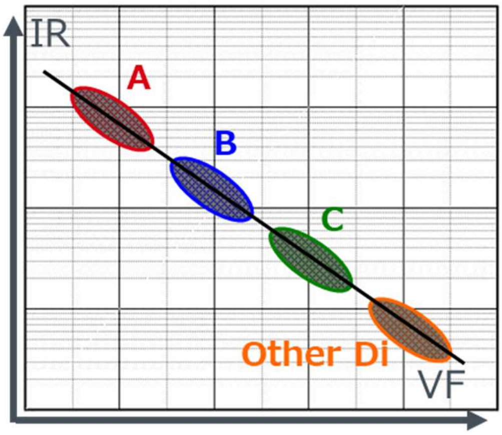


Figure 10. Differences in IR-VF characteristics due to different barrier metals

1.3.2 Thermal Runaway of Si SBDs

Thermal runaway is an important consideration when using Si SBDs. This is a phenomenon in which the Tj of a diode exceeds its maximum rating and, in the worst case, can lead to its destruction. As mentioned earlier, the IR dissipation of Si SBDs cannot be ignored. The heat generation is the product of VR and IR, i.e., the power loss due to reverse voltage and reverse current flowing multiplied by the thermal resistance. This is the same as the usual heat generation calculation. In the example shown here, SBD using metal A with the highest IR is the most disadvantageous. IR tends to increase as VR increases (Figure 9) and has a positive temperature characteristic that increases as temperature increases (see Figure 7). From these characteristics, thermal runaway may occur due to an increase in Tj caused by self-heating (or by a rise in the ambient temperature), causing the IR value to increase, leading to further heat generation and a further rise in IR. In order to prevent thermal runaway, the thermal design must be adequate to enable heat dissipation even when heat is generated under various conditions. Below are a few important points relating to thermal runaway.

1.3.3 Conceptual Images of Applications for Different VF and IR

Figure 11 shows an image of suitable applications according to the VF and IR characteristics of Si SBDs. We hope you will find it useful as a reference when selecting components for your design. However, it is necessary to consider the suitability under the design specifications and actual use conditions.

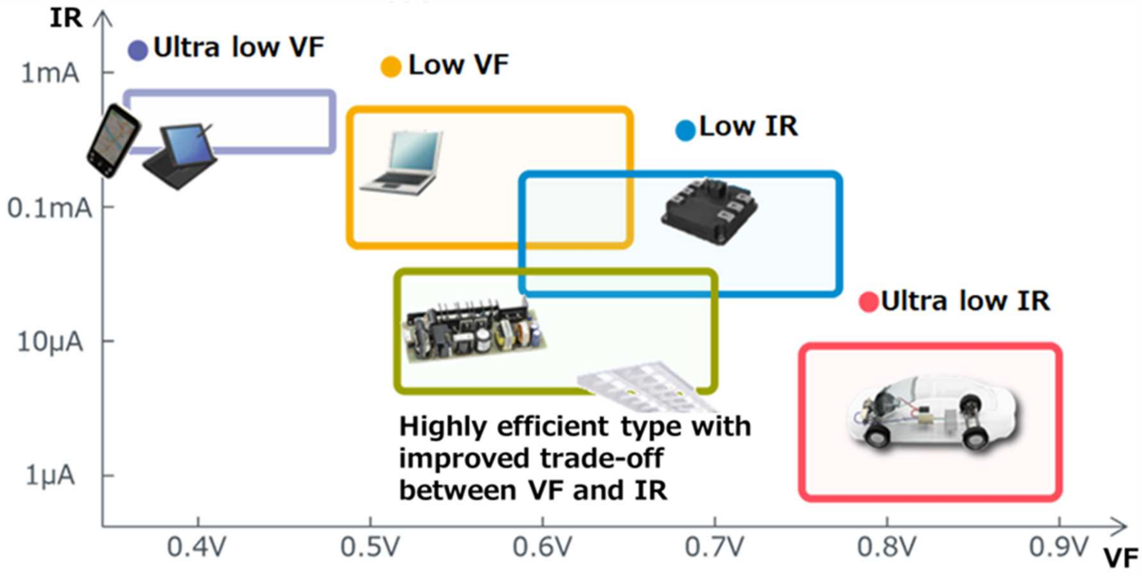


Figure 11. Conceptual images of applications for different VF and IR

<Summary>

- Si SBD characteristics differ depending on the barrier metal.
- It should be recognized that IR values of Si SBD are sufficiently large that they cannot be ignored.
- Thermal runaway remains a real possibility, so the thermal design should be verified adequately.

1.4 Si Fast Recovery Diodes (FRD)

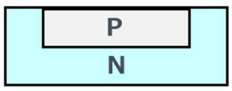
The following is a description of the features and characteristics of FRDs and their improvement, as well as application illustrations.

1.4.1 Features of FRDs

FRDs are Si PN junction diodes, characterized by high trr speed and generally high VF. For example, the VF for a general-purpose 10 A-class FRD is a little under 2 V. This is because of the tradeoff between a rapid-acting trr and the VF value. However, devices with a greatly reduced VF are also being developed, and the optimal FRD can be selected according to the application.

Table 3 summarizes characteristics of FRDs types and applications suitable for each. In the table, the symbol “×” should be understood to mean inferior or poor compared with other devices.

	VF	IR	trr	High Temp.	Suitable application
Ultra fast	×	◎	○	○	PFC (Continuous current mode)
Fast, Low VF (balanced)	△	○	○	○	General purpose
Ultra low VF	○	△	○	○	PFC (Boundary current mode)
Other diodes	◎	×	○	○	Diode bridge



* FRD characteristics differ depending on impurities diffused in the chip

Table 3. FRDs: Characteristics of types and suitable applications

Ultra-high-speed devices have high VF values, but because IR is low, they experience low losses in continuous current mode of PFC (power factor correction) applications, to which they are well suited.

FRDs that balance fast operation with a low VF are extremely versatile, and the speed of FRDs can be utilized in diverse applications.

The IR characteristic of ultra-low VF devices is somewhat inferior, but conduction losses can be reduced due to its very low-VF characteristics, making such diodes suited to boundary current mode PFC applications with large peak currents.

The graphs in Figure 12 illustrate the tradeoff between VF and trr. The device represented by the orange curves is an FRD with a low VF but a long trr, and the IR during this time is also large. The device represented by the red curves has the opposite characteristics. We see that faster devices have a high VF, whereas lower VFs mean that trr/IR is large.

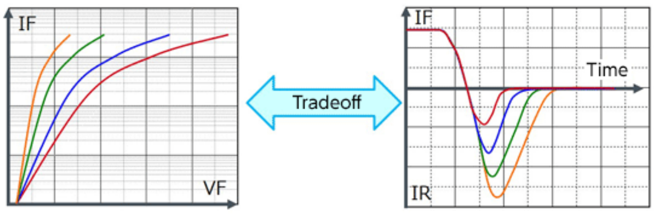


Figure 12. FRD trr is a tradeoff between the magnitude of VF

1.4.2 Noise of FRDs

From the standpoint of EMC, noise originating in switching power supplies and other switching phenomena is an important matter for study. The fast trr of FRDs means that noise occurs during reverse recovery, that is, during the trr period. Figure 13 illustrates noise during reverse recovery and after improvement. IRp represents the peak of the reverse current when the FRD turns off. The recovery slope or gradient is indicated by dir/dt. Noise during reverse recovery can be reduced if the IRp can be made small and the dir/dt can be kept gentle.

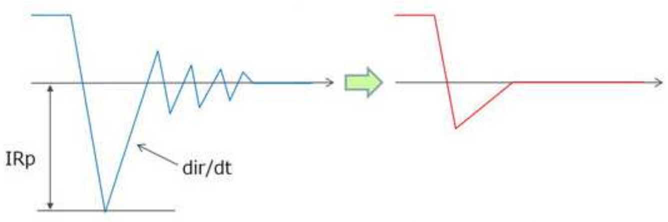


Figure 13. Noise at reverse recovery of FRD and image after improvement

Such low-noise FRDs are called "soft recovery" types. In actual development, it was necessary to take different remedial measures to reduce IRp and moderate dir/dt. Initially the impurity concentration of the P-type silicon (anode) was reduced in order to lower the IRp. By this means the leak current was reduced and IRp also fell, but dir/dt was still steep and noise remained, and there was the further problem of a higher VF, resulting from the tradeoff (Figure 14, top).

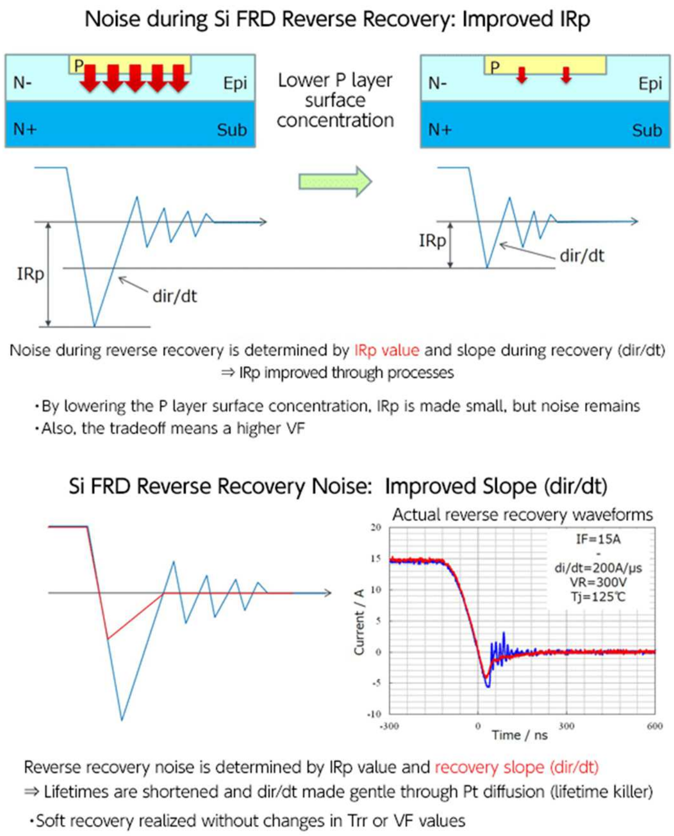


Figure 14. Improvements of noise at reverse recovery of FRD

Next, in order to make the slope gentler, platinum (Pt) was diffused as a “lifetime killer” to shorten the lifetime, and dir/dt was successfully decreased. The lifetime is the time in the PN junction from when the off/reverse-bias state is entered until there is a return to the original state through recombination of minority carriers. If this lifetime is long, the residual carriers cause phenomena such as the persistence of currents. A “lifetime killer” hastens the recombination to shorten the lifetime. In this case, impurities are diffused as lifetime killers; in this case, Pt was used.

We turn to production processes. This may be a bit difficult for persons who are not familiar with production, but based on these two methods, ultimately FRDs were developed having greatly reduced reverse recovery noise, while maintaining the trr and VF characteristics essentially unchanged. The actual waveforms shown in Figure 14 bottom right compare the soft recovery type (red) and standard type (blue) devices. Clearly the noise level is extremely low for the soft recovery type.

1.4.3 Applications of FRDs Drawing on trr and VF

Figure 15 illustrates the characteristics of FRDs and suitable applications. In the hope that this will be of use in design. As explained previously, devices with a fast trr have a high VF, and those with a low VF have slower trr; this basic tendency is considered when selecting devices suited to application characteristics.

Where PFC is concerned, devices with extremely low VF values in BCM (boundary current mode), and devices with extremely fast trr values in CCM (continuous current mode), each contribute to lessen losses.

Moreover, devices with reduced noise and with an optimized tradeoff between VF and trr are suited to use in an extremely broad range of applications.

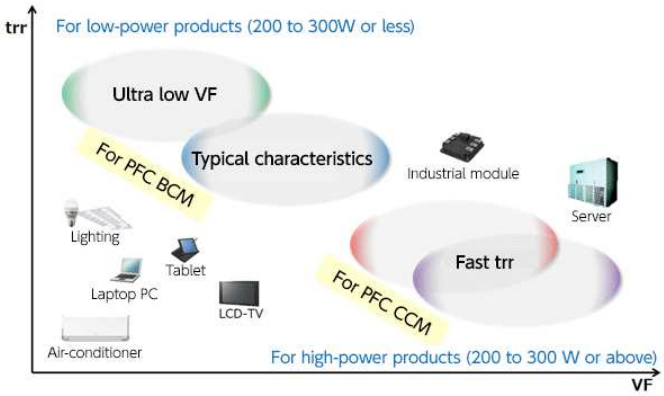


Figure 15. Applications of FRDs drawing on trr and VF

<Summary>

- FRD characteristics differ depending on the impurities diffused in the silicon.
- There is a tradeoff between the VF and trr values of FRDs.
- Noise during reverse recovery has adverse effects in switching power supply applications, and so improved versions are being developed.

2. Si Transistors

Following diodes, we will discuss transistors, which are representative discrete semiconductors, and Si transistors can be categorized by manufacturing process and structure, such as bipolar transistor and MOSFET. Si transistors can also be categorized according to the currents and voltages they can handle and their application areas. Here we are focusing on power devices, and so among the wide variety of transistors in use, we consider only power transistors. Among these, we intend to primarily address MOSFETs, which in recent years have been widely used in applications to control high power levels.

2.1 Categories of Si Transistors

When considering Si transistor categories, there are a number of methods of categorization depending on the criteria used. Here, although somewhat imprecise, we have categorized them in terms of device structures and processes, as shown in Figure 16. Of these, Si transistors which are relevant to our current theme of power devices are indicated in a bold font with colored background.

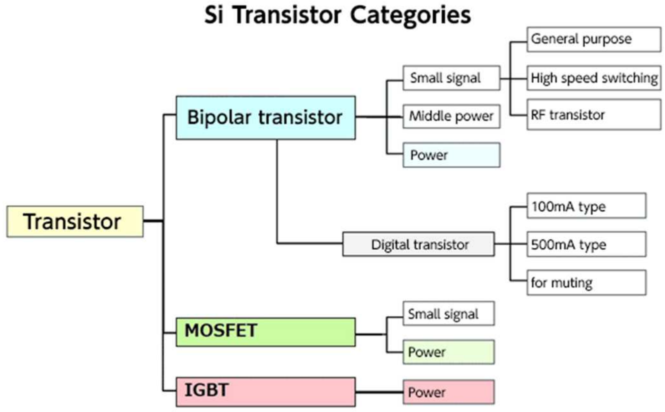


Figure 16. Example of Si transistor categories from structure and process aspects

With bipolar transistors and MOSFETs, there are power type and small signal type. IGBTs are transistors that were originally developed to handle large amount of power, however, and so are essentially power devices. In passing we note that MOSFET is an acronym for Metal Oxide Semiconductor Field Effect Transistor, and so these are one type of FET (field effect transistor). IGBT is an acronym for Insulated Gate Bipolar Transistor.

2.2 Features of Si Transistors

Below we summarize the features of the major performance areas of transistors, for Si bipolar transistors, MOSFETs, and IGBTs in Tabel 4. Evaluations of various items are based on representative characteristics of each transistor, and so there may not be consistency among separate transistors. These should be regarded as overall tendencies and features. The structure, principle of operation, and major characteristics of each transistor are explained individually in the following sections.

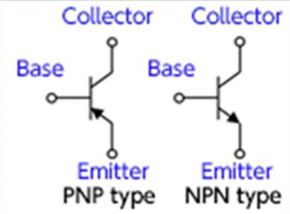
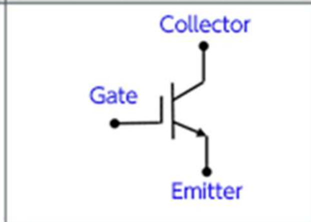
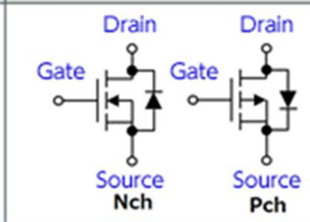
	Bipolar transistor	IGBT	MOSFET
Symbol	 Collector Collector Base Base Emitter Emitter PNP type NPN type	 Collector Gate Emitter	 Drain Drain Gate Gate Source Source Nch Pch
Drive	(Complicated) Current driven	(Easy to drive) Voltage driven	(Easy to drive) Voltage driven
Driving power	Somehow high	Low	Low
Switching time	Low (electron+hole)	Low (electron+hole)	Low (electron+hole)
Temperature stability	Rising h_{FE} · Falling V_{BE}	High stability	High stability
Higher voltage	Easy	Easy	Requires structure modification

Table 4. Main features and comparison of Si transistors

2.2.1 Si Bipolar Transistors: Structure, Operation Principle, and Major Characteristics

Figure 17 is a schematic diagram of an NPN-type bipolar transistor. Bipolar transistors are constructed with a silicon PN junction, and current flows between the collector and emitter when current is applied to the base. As shown in Table 5, with respect to the drive, the base current is adjusted according to the relationship between the amplification factor and the collector current; the major difference from a MOSFET is that the bias (base) current for amplification or on/off flows through the transistor base.

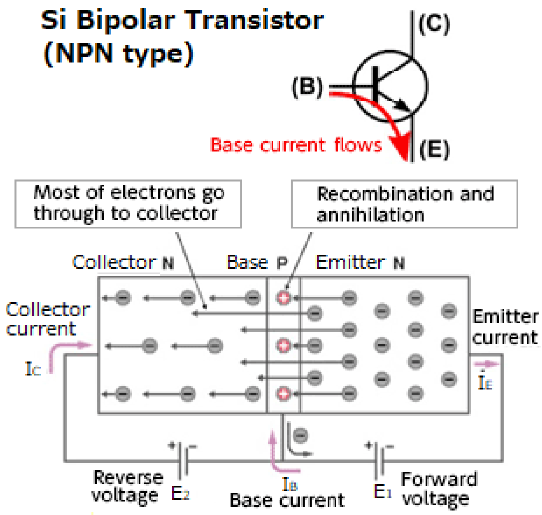


Figure 17. Si bipolar transistor structure and principle of operation

VCEO [V]	**Absolute maximum rating** for the voltage that can be applied across collector and emitter with base open
IC [A]	**Maximum collector current.** The maximum value of the current that can be passed to the collector.
PC [W]	**Maximum collector loss.** Maximum value of the product of collector-emitter voltage VCE and collector current IC; the maximum value of the power that can be applied to the transistor
hFE	**Direct current amplification factor.** hFE = IC/IB. Indicates the extent to which the collector current amplifies the (DC) base current.
VCE(sat) [V]	**Maximum value of the collector-emitter voltage** under conditions of arbitrary base current and collector current values
S.O.A (A.S.O)	**Area of safe operation.** Range of operation over which the transistor can be used without failure or degradation

Table 5. Major Characteristics of Si Bipolar Transistors

2.2.2 Si MOSFETs: Structure, Operation Principle, and Major Characteristics

Figure 18 shows a schematic diagram of an N-channel MOSFET. A MOSFET conducts by forming a channel between the source and drain when a voltage is applied to the gate. Since the gate is insulated from the source and drain by an oxide film, no current flows in the sense of conduction, but a charge called Qg (see Table 6) is required to drive the gate.

MOSFETs have a parameter called the on-resistance RDS(on), which is particularly important when handling large amounts of power (see Table 6). Bipolar transistors, however, do not have an on-resistance. The world's first transistors were bipolar devices, and so the order may appear to be reversed, but in recent years MOSFETs have been the mainstream for power supply circuits in particular, and I think many people will have first used MOSFETs, and so we will concentrate mainly on these devices. To return from our digression, the VCE (sat), called the collector-emitter saturation voltage, corresponds to the on-resistance of a bipolar transistor. When a predetermined collector current is passed, that is, when the transistor is turned on, there is a voltage drop, and the resistance while turned on can be determined from this voltage drop.

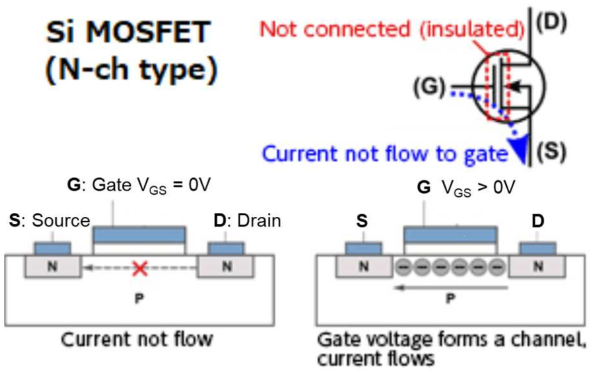


Figure 18. Si MOSFET structure and principle of operation

VDSS[V]	**Drain-source voltage.** Maximum allowable voltage that can be applied between drain and source.
VGS(th)	**Threshold voltage.** Gate voltage at which the device is turned ON.
RDS(on)	**Drain-source resistance value** with the device turned ON
PD[W]	**Allowable loss.** Maximum allowable power dissipation. depending on Ta and Tc.
ID[A]	**Maximum allowable current value that can flow to drain,** determined by on-resistance and power loss.
Ciss, Coss, Crss	Capacitance between each pin, affecting switching speed.
tr,td(on) tf,td(off)	**On/Off time,** depending on current.
Qg, Qgs, Qgd	**Amount of charges required to turn on,** affecting switching speed.
S.O.A (A.S.O)	**Area of safe operation.** Range of operation over which the transistor can be used without failure or degradation

Table 6. Major Characteristics of Si MOSFETs

2.2.3 Si IGBTs: Structure, Operation Principle, and Major Characteristics

IGBTs have a structure that is a hybrid of bipolar transistors and MOSFETs (Figure 19). They were developed in order to exploit the advantages of both MOSFETs and bipolar transistors. Like MOSFETs, under gate voltage control they are capable of fast operation, but they combine this with the low on-resistance even for high voltages of bipolar transistors.

Operation is similar to that of MOSFETs: by applying a voltage to the gate, a channel is formed, and current flows. Whereas in a MOSFET (an N channel type example) current flows between the source and drain, which are the same N type, an IGBT has a structure in which current flows from a P-type collector to an N-type emitter, similar to the structure of a bipolar transistor. Hence an IGBT has the gate-related parameters of a MOSFET, and the collector-emitter related parameters of a bipolar transistor (see Table 7).

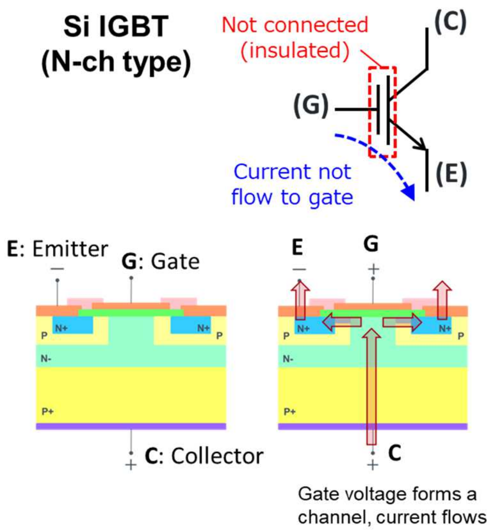


Figure 19. Si IGBT structure and principle of operation

VCES[V]	**Absolute maximum rating for collector-emitter voltage**
IC[A]	**Maximum collector current.** The maximum value of the current that can be passed to collector.
P[W]	**Maximum collector loss.** Maximum value of the product of collector-emitter voltage VCE and collector current IC; the maximum value of the power that can be applied to the transistor
VCE(sat)[V]	**Maximum value of collector-emitter voltage** under conditions of an arbitrary gate voltage and collector current values
Ciss,Coss,Crss	**Capacitance between each pin,** affecting switching speed.
tr,td(on) tf,td(off)	**On/Off time,** depending on current.
Qg, Qge, Qgc	**Amount of charges required to turn on,** affecting switching speed.
S.O.A (A.S.O)	**Area of safe operation.** Range of operation over which the transistor can be used without failure or degradation

Table 7. Major Characteristics of Si IGBTs

2.2.4 Comparison of Si Transistor Basic Operation Characteristics

Three types of transistors mentioned above have different operation characteristics. Figure 20 shows the basic operating (current-voltage) characteristics. Since power devices are basically used as switches, the lowest possible voltage (Vce/Vds) is more advantageous from a loss perspective, even if current (Ic/Id) flows when the device is on. This is one representative characteristic that is used when determining which transistor is optimal for the circuit conditions of the application.

3. Si MOSFETs

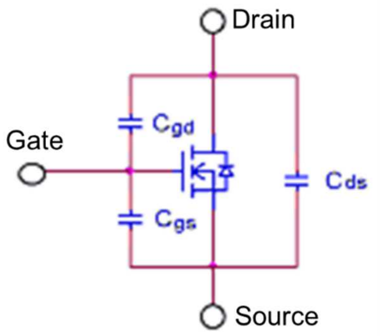
The following is a more detailed description of the characteristics of Si MOSFETs, which are widely used as power switches today.

3.1 Parasitic Capacitance of Si MOSFETs

Due to their structure, MOSFETs have a parasitic capacitance, as indicated in Figure 21. It is for an example of an N-channel MOSFET, but the situation is much the same for P-channel MOSFETs. In the power MOSFETs that handle large amounts of power, the parasitic capacitance must be regarded as a parameter that limits the usage frequency and switching speed.

The drain and source of a MOSFET are insulated from the gate by the gate oxide film. A PN junction is formed between the drain and source with substrate intervening, and a parasitic ("body") diode is present.

The gate-source capacitance Cgs and gate-drain capacitance Cgd in the diagram below are determined by the capacitance of the gate oxide film. The drain-source capacitance Cds is the junction capacitance of the parasitic diode.



Symbol	Expression	Meaning
Ciss	Cgs+Cgd	Input capacitance
Coss	Cds+Cgd	Output capacitance
Crss	Cgd	Feedback capacitance

Figure 21. Parasitic capacitance of Si MOSFETs

The three parameters Ciss, Coss, Crss appearing on MOSFET data sheets in general relate to these parasitic capacitances. On data sheets which provide separate descriptions of static characteristics and dynamic characteristics, these are classified as dynamic characteristics. These are important parameters affecting switching performance.

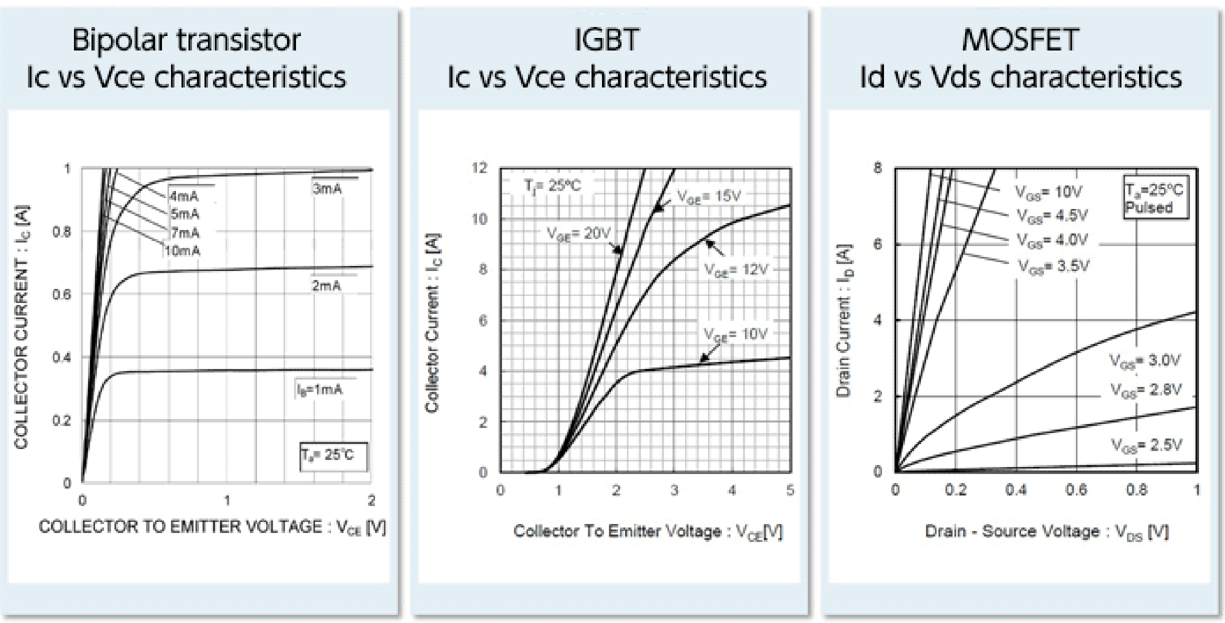


Figure 20. Basic operation characteristics of Si transistors

Ciss is the input capacitance, and is the capacitance obtained by totaling the gate-source capacitance Cgs and the gate-drain capacitance Cgd; it is the capacitance of the MOSFET as a whole, as seen from the input. This capacitance must be driven (charged) in order to cause the MOSFET to operate, and so is a parameter of importance when studying the drivability of an input device or power consumptions. Qg is the amount of charge necessary to drive (charge) Ciss.

Coss is the output capacitance, obtained by adding the drain-source capacitance Cds and the gate-drain capacitance Cgs, and is the total capacitance on the output side. If Coss is large, a current arising due to Coss flows at the output even when the gate is turned off, and time is required for the output to turn off completely.

Crss is the gate-drain capacitance Cgd itself, and is called the feedback capacitance or the reverse transfer capacitance. If Crss is large, the rise in drain current is delayed even after the gate is turned on, and the fall in current is delayed after the gate is turned off. In other words, this parameter greatly affects switching speed. Qgd is the charge amount necessary to drive (charge) Crss.

These capacitances exhibit a dependence on the drain-source voltage V_{DS}. As indicated in the graph of Figure 22, there is a tendency for capacitance values to be reduced as V_{DS} is increased.

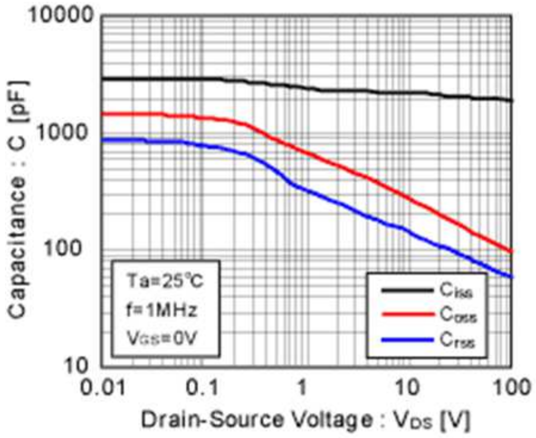


Figure 22. Dependence of Si MOSFET parasitic capacitance on V_{DS}

3.1.1 Temperature characteristic of Si MOSFET parasitic capacitances

Ciss, Coss and Crss as the MOSFET parasitic capacitances change hardly at all with temperature. Hence it can be said that switching characteristics are hardly affected at all by temperature changes. Actual measurement examples of each parasitic capacitance are shown in Figure 23, Figure 24, and Figure 25.

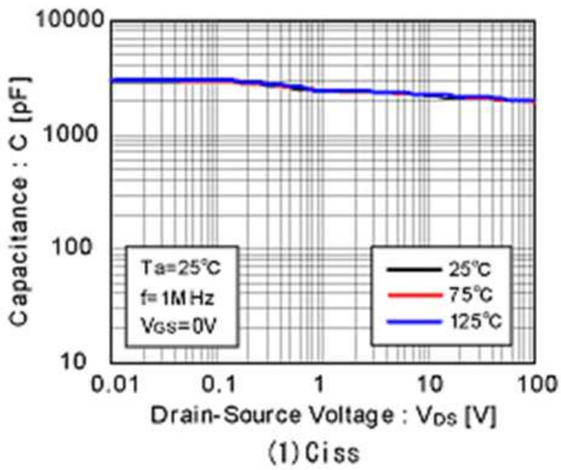


Figure 23. Temperature characteristics of Si MOSFET parasitic capacitance Ciss

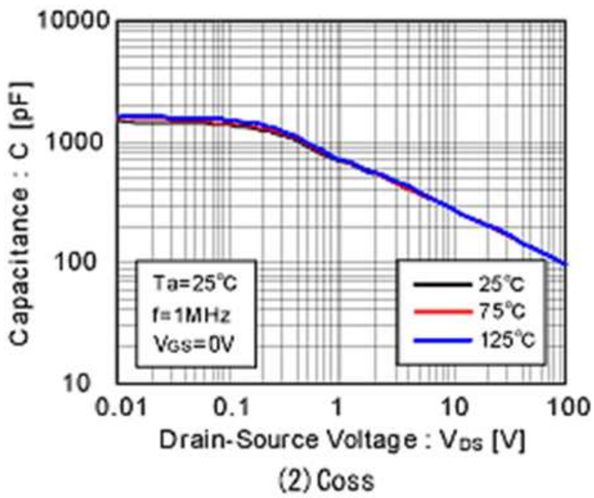


Figure 24. Temperature characteristics of Si MOSFET parasitic capacitance Coss

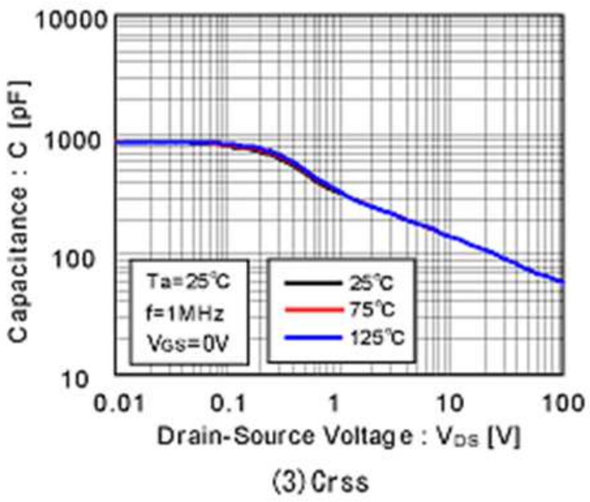


Figure 25. Temperature characteristics of Si MOSFET parasitic capacitance Crss

3.2 Switching Characteristics of Si MOSFETs

In power conversion, a MOSFET is essentially used as a switch. As MOSFET switching characteristics, the turn-on delay time $t_{d(on)}$, rise time t_r, turn-off delay time $t_{d(off)}$, and fall time t_f are generally indicated. Figure 26 is excerpted from the datasheet for the N-ch 600V 4A MOSFET: R6004KNX, which features a low turn-on resistance and fast switching. The names and symbols of these parameters may differ somewhat among manufacturers. For example, the turn-on/off delay time may be called the on/off delay time, and the rise/fall time may be the rising/falling time, and so on.

● Electrical Characteristics (T_a = 25℃)

Parameter	Symbol	Conditions	Values			Unit
			Min.	Typ.	Max.	
Turn-on delay time	$t_{d(on)}$ *5	V_{DD} ≈ 300V, V_{GS} = 10V	-	15	-	ns
Rise time	t_r *5	I_D = 2A	-	10	-	
Turn-off delay time	$t_{d(off)}$ *5	R_L ≈ 150Ω	-	30	-	
Fall time	t_f *5	R_G = 10Ω	-	25	-	

Figure 26. Excerpt from Si MOSFET datasheet specification table

These parameters related to switching are greatly affected by the signal source impedance and drain load resistance R_L of the measurement circuit. Hence, reference measurement conditions are stipulated and a measurement circuit is suggested. In general, the conditions for V_{DD}, V_{GS}, I_D, R_L, and R_G are shown in the Conditions in Figure 26, and for measurement, the presented measurement circuit (Figure 27) is used, the specified conditions are applied, and measurements are made based on the parameter definitions (Figure 28). The parameter definitions in Figure 28 are expressed in text as follows.

- $t_{d(on)}$ Time from 10% of the rise of V_{GS} until 10% of the rise of V_{DS}
- t_r Time from 10% to 90% of the rise of V_{DS}
- $t_{d(off)}$ Time from 90% of the fall of V_{GS} until 90% of the fall of V_{DS}
- t_f Time from 90% to 10% of the fall of V_{DS}

Here, the expression "rise/fall" of V_{DS} may seem to be the opposite when looking at the waveforms, but the output is inverted, so this is how it is expressed.

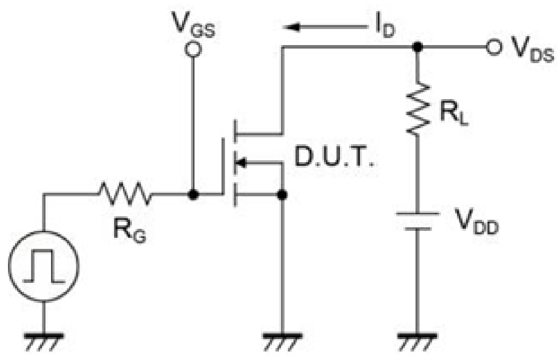


Figure 27. Measuring circuit for the parameters in Figure 26

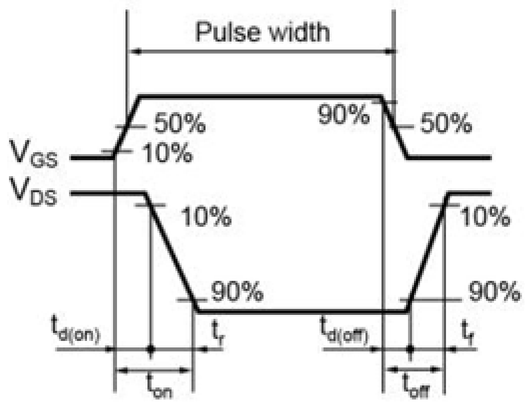


Figure 28. Definition of parameters in Figure 26

3.2.1 Temperature Characteristics of Si MOSFET Switching Characteristics

The parameters (time) for MOSFET switching tend to increase slightly as the temperature rises, but since a temperature increase of 100℃ results in a switching time increase of about 10%, the temperature dependence can be thought of as almost nil. Actual measurement examples are shown in Figures 29 through 32.

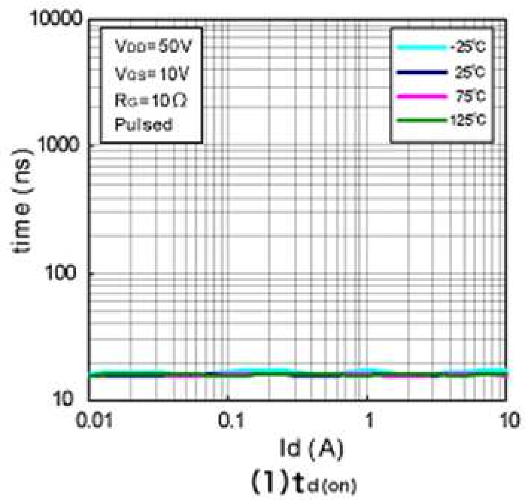


Figure 29. Example of measured turn-on delay time $t_{d(on)}$

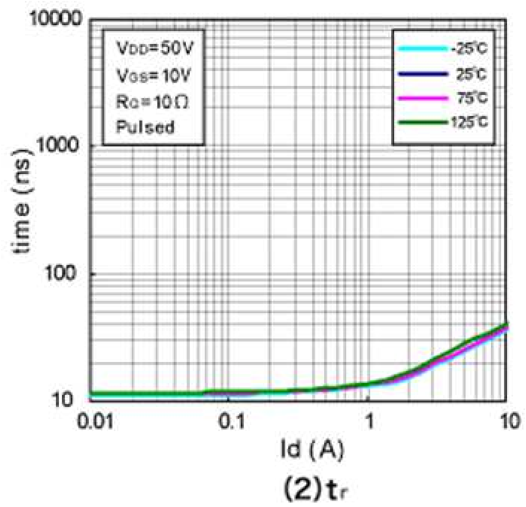


Figure 30. Example of measured rise time t_r

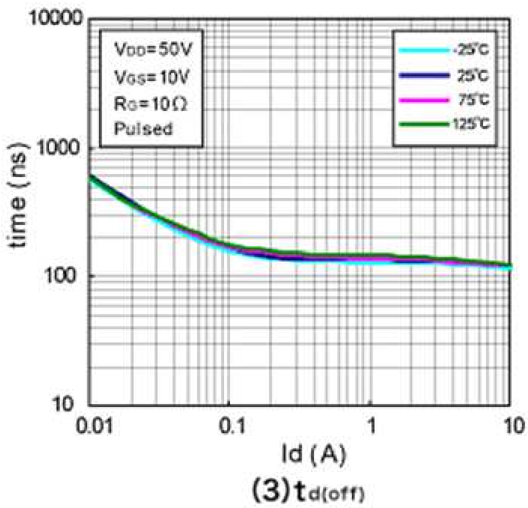


Figure 31. Example of measured turn-off delay time $t_{d(off)}$

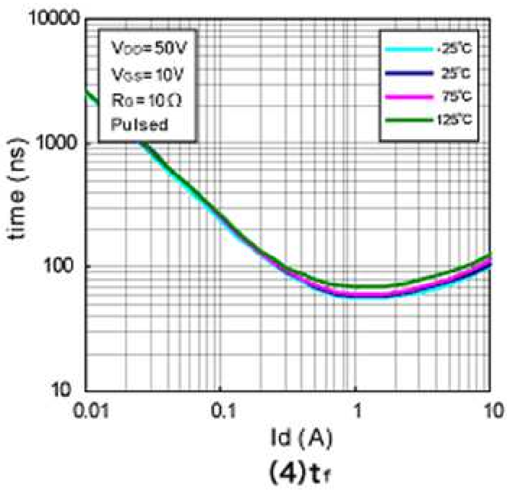


Figure 32. Example of measured fall time t_f

<Summary>

- MOSFET switching characteristics that are generally provided include the turn-on delay time, rise time, turn-off delay time, and fall time.
- Switching characteristics are greatly affected by the measurement conditions and measurement circuit, and so the suggested conditions should be confirmed.
- Switching characteristics are affected hardly at all by changes in temperature.

3.3 Si MOSFET Gate Threshold Voltage

The gate threshold voltage $V_{GS(th)}$ of a MOSFET is the voltage required between the gate and source to turn on the MOSFET. In other words, if V_{GS} is greater than or equal to $V_{GS(th)}$, the MOSFET will turn on. Figure 33 is an excerpt from the datasheet of an N-ch 600V/4A MOSFET: R6004KNX.

●**Electrical characteristics** (T_a = 25°C)

Parameter	Symbol	Conditions	Values			Unit
			Min.	Typ.	Max.	
Drain - Source breakdown voltage	$V_{(BR)DSS}$	V_{GS} = 0V, I_D = 1mA	600	-	-	V
Zero gate voltage drain current	I_{DSS}	V_{DS} = 600V, V_{GS} = 0V				μA
		T_j = 25°C	-	-	100	
		T_j = 125°C	-	-	1000	
Gate - Source leakage current	I_{GSS}	V_{GS} = ±20V, V_{DS} = 0V	-	-	±100	nA
Gate threshold voltage	$V_{GS(th)}$	V_{DS} = 10V, I_D = 1mA	3	-	5	V
Static drain - source on - state resistance	$R_{DS(on)}$ *5	V_{GS} = 10V, I_D = 1.5A				Ω
		T_j = 25°C	-	0.90	0.98	
		T_j = 125°C	-	1.36	-	
Gate resistance	R_G	f = 1MHz, open drain	-	3.3	-	Ω

Figure 33. Example of Si MOSFET gate threshold voltage specification

The blue line surrounds information on $V_{GS(th)}$, and the column of Conditions indicates that conditions are V_{DS} = 10 V and I_D = 1 mA. Under these conditions, and at Ta = 25℃, $V_{GS(th)}$ is guaranteed to have minimum and maximum values of 3 V and 5 V. In other words, as V_{GS} is raised, the MOSFET begins to turn on (I_D begins to flow), and when I_D is 1 mA, the value of V_{GS} is between 3 V and 5 V inclusive; this value is $V_{GS(th)}$.

3.3.1 Si MOSFET Gate Threshold and I_D-V_{GS} Characteristics

Figure 34 is a graph of I_D-V_{GS} characteristics of the example MOSFET. From this graph, $V_{GS(th)}$ can be read out: V_{DS} is 10V, the same as in the specifications table in Figure 33, so $V_{GS(th)}$ is the V_{GS} when I_D is 1mA,

and so the V_{GS} when the curve for T_a = 25℃ intersects the I_D = 1 mA (0.001 A) line is approx. 3.8 V. The datasheet (Figure 33) does not indicate a representative or typical value (indicated by “Typ.”), and we see from the graph that the Typ. value for $V_{GS(th)}$ is about 3.8 V. The graph value can in essence be regarded as the Typ. value.

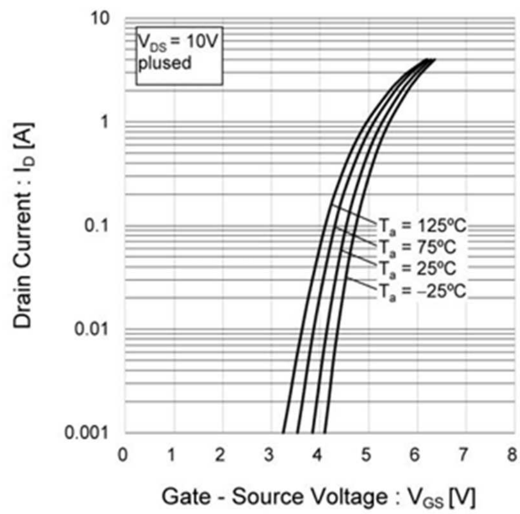


Figure 34. Example of Si MOSFET I_D-V_{GS} characteristics

$V_{GS(th)}$ as the specification value is V_{GS} when I_D = 1mA, but in actual use of 4A MOSFET, it is unlikely to use I_D of 1mA. For example, when an I_D of 1 A at Ta = 25℃ is required, we see from the graph that the V_{GS} is about 5.3 V. In actual design, the appropriate V_{GS} must be confirmed from the I_D-V_{GS} characteristics.

3.3.2 Temperature Characteristics of Si MOSFET Gate Threshold and I_D-V_{GS} Characteristics

The I_D-V_{GS} characteristic graph in Figure 34 also shows temperature characteristics. From the graph, it can be read that I_D tends to increase at higher temperatures if V_{GS} is constant. In the previous example of a V_{GS} of 5.3V read from a condition of Ta=25℃ and I_D=1A, I_D increases to about 1.5A at Ta=75℃, so care is required depending on the operating conditions.

Next, Figure 35 shows the temperature characteristic graph of $V_{GS(th)}$ itself. As with the value read from the I_D-V_{GS} characteristic in Figure 34, $V_{GS(th)}$ at 25 ℃ is approximately 3.8V. The temperature in this graph is T_j, but both Figure 34 and Figure 35 have "pulsed" in the conditions, which indicates that the data is from a pulse test, so it is safe to assume that $T_j \approx T_a \approx$ 25℃.

As for temperature characteristics, we see that $V_{GS(th)}$ tends to decrease at higher temperatures. In other words, it means that the same I_D can flow at lower V_{GS}. Naturally, this is consistent with the temperature characteristics of I_D-V_{GS} in Figure 34.

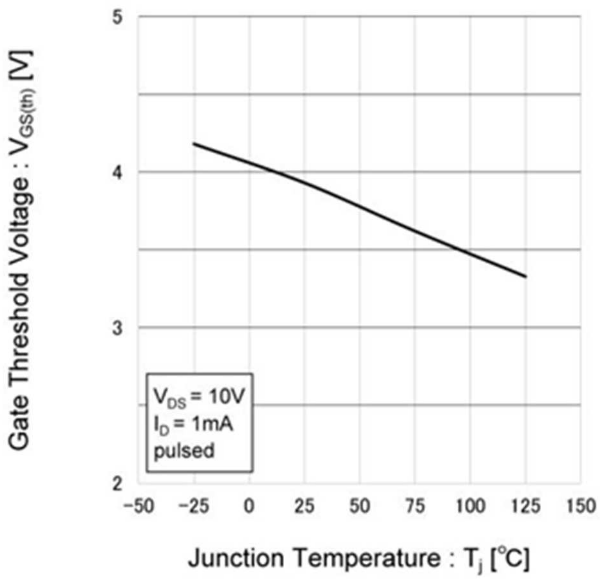


Figure 35. Example of Si MOSFET $V_{GS(th)}$ temperature characteristics

<Summary>

- The voltage at which the MOSFET turns on is called the gate threshold $V_{GS(th)}$.
- If V_{GS} is constant, a rise in temperature will cause I_D to increase, and so conditions of use must be considered carefully.

3.4 Super-Junction MOSFETs

The following is an example of a Si MOSFET to illustrate its features, and a recent high-voltage representative among Si MOSFETs, the super junction MOSFET (hereafter SJ MOSFET), will be discussed.

3.4.1 Features and Positioning of Power Transistors

Before going into the explanation of SJ MOSFETs, we would like to review the power and frequency coverage of Si MOSFETs, IGBTs, and SiC MOSFETs, which are the major power transistors in recent years. If you understand the position of Si MOSFET among power transistors and can imagine how to use them based on their features and characteristics, it will be easier to understand the explanation of SJ MOSFETs that follows.

Figure 36 shows the approximate power and frequency range that each power transistor can handle, and although Si MOSFETs are a step behind IGBTs and SiC MOSFETs in this comparison in terms of on-resistance Ron and breakdown voltage ("Voltage" in the radar chart), they are in a superior position for high-speed operation at low to medium power.

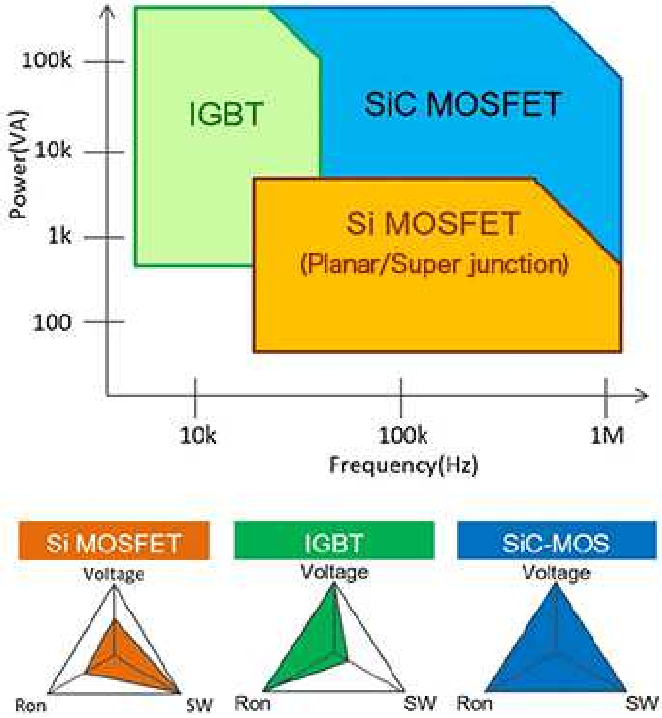


Figure 36. Power and frequency coverage images of major power transistors

3.4.2 Planar MOSFETs and SJ MOSFETs

Si MOSFETs can be classified as planar MOSFETs and SJ MOSFETs according to the manufacturing processes used. For background, in the field of power transistors, the super-junction structure was developed in order to transcend the limits of planar structures.

As indicated in Figure 37, a planar structure constitutes a flat or planar transistor. This structure has had the drawback that if the rated voltage is raised, the drift layer becomes thicker, and so the on-resistance is increased. In contrast, a super-junction structure is a structure in which multiple vertical pn junctions are arranged, as a result of which a low on-resistance $R_{DS(ON)}$ and reduced gate charge Qg are realized while maintaining a high voltage.

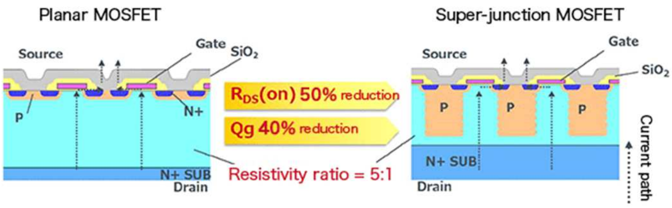


Figure 37. Structural Comparison of Planar MOSFETs and SJ MOSFETs

In addition, the reverse current irr and the reverse recovery time trr of the internal diode are parameters that need to be studied for the turn-off switching characteristics of a transistor. As indicated in Figure 38 waveform diagram, in essence a SJ MOSFET has a larger pn junction area than a planar MOSFET, and so trr is faster than for a planar MOSFET, but a larger irr flows. However, this is a challenge for SJ MOSFETs that is being improved. In addition, there are variations among SJ MOSFETs based on features such as high speed and low noise.

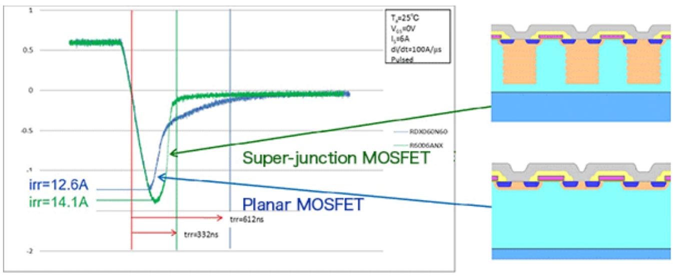


Figure 38. Off-switching characteristics of planar MOSFETs and SJ MOSFETs

<Summary>

- Si MOSFETs are in a position among power transistors to operate at high speeds with low to medium power.
- Super-junction structure reduces on-resistance $R_{DS(ON)}$ and gate charge Qg while maintaining breakdown voltage compared to planar structure.
- SJ MOSFETs have characteristics of faster trr but larger irr than planar MOSFETs, but improvements are being made.

4. Confirming Transistor Suitability in Actual Operation

In circuit design, transistors that can meet the requirements of the circuit are usually selected by referring to the specifications on the data sheet, and a prototype is made to confirm actual operation. In the prototyping, it is necessary to check the operation and phenomena that cannot be verified from the verification of the schematic on the desk, and to judge whether the selected transistor is appropriate or not. Chapter 4 describes the procedures and methods to comprehensively examine the selected transistors from both electrical and thermal perspectives to ensure that they will perform well in actual operation.

4.1 Confirmation Flow for Transistor Suitability in Actual Operation

From this point on, the procedure for determining whether the selected transistor has a problem in actual operation is explained .An example of the procedure is shown in the flowchart in Figure 39.

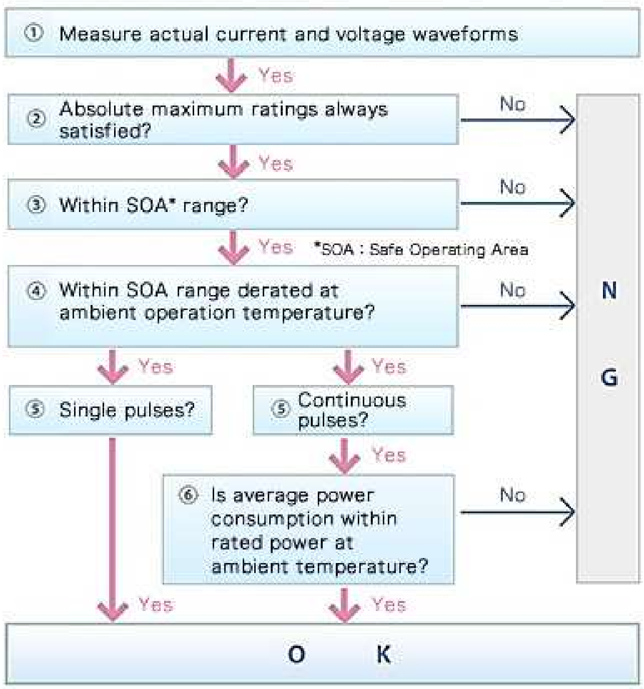


Figure 39. Confirmation flow for transistor suitability in actual operation

The following explanations follow the flowchart ① through ⑥.In the explanation, "Continuous pulse" is selected for ⑤ because a switching circuit is assumed. Also, although it is not shown in the chart, "temperature check of transistor chip" is added at the end as ⑦ just in case.

4.1.1 ①Measurement of Actual Current and Voltage Waveforms

Flowchart ①. First, it is necessary to identify what kind of voltage and current the transistor is handling in the circuit operation. Here we use a switching circuit as an example. The transistor is a low-noise SJ MOSFET: EN series R6020ENZ (600V, 20A, 0.17Ω, TO-3PF) as an example.

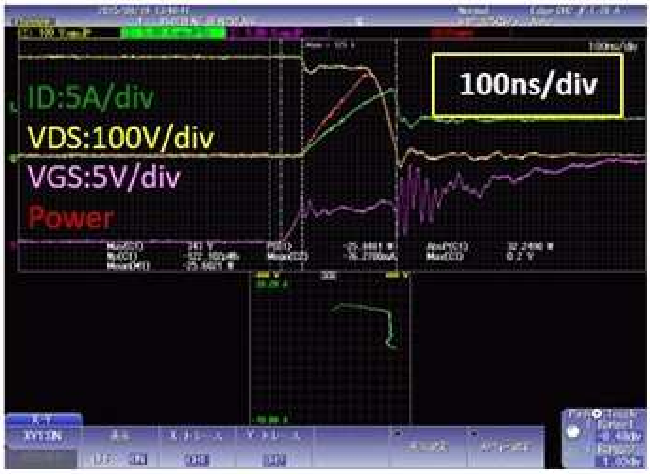
Specific measurements include taking voltage and current waveform data using an oscilloscope. For operating conditions, it is customary to use worst-case conditions under operating conditions such as input/output voltage, load current, and temperature. The waveforms in Figure 40 show the entire switching operation of a MOSFET.

In addition to this, enlarged waveform data during on-off transitions is also captured in order to calculate power losses during switching, explained below. Here, we have also captured the waveforms for I_D, V_{DS}, V_{GS}, and the power, shown in Figure 41. Waveforms during the on and off states, and during transitions between these states, are needed.

<Summary>

- In prototyping, it is essential to determine whether a selected transistor can be used in actual operation.
- For purposes of confirmation, the voltages and currents handled by the transistor are measured.

OFF to ON transition



ON to OFF transition

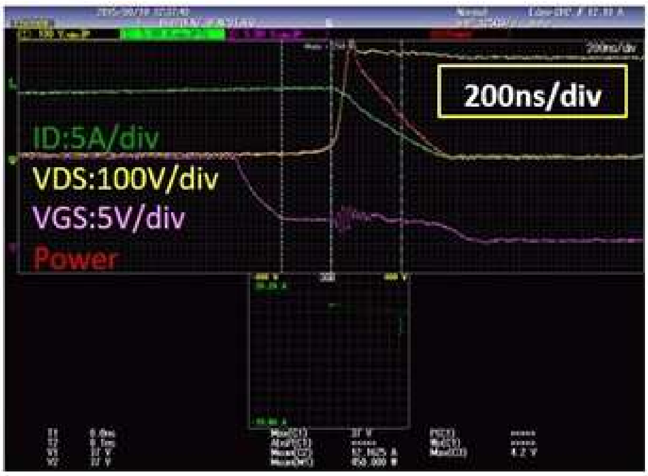


Figure 41. Magnified waveform during MOSFET on/off transition

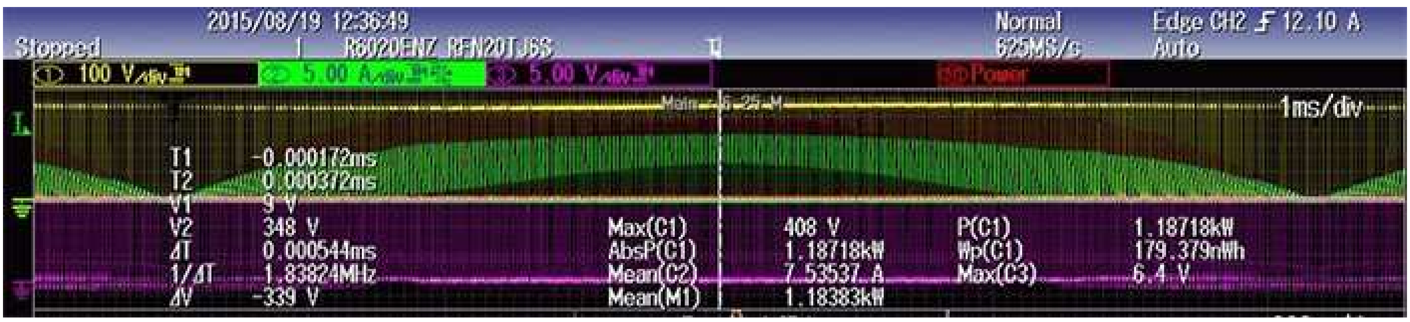


Figure 40. Overall view of MOSFET switching operation waveform

4.1.2 ② Confirmation of Being within Absolute Maximum Ratings

Flowchart ②. Absolute maximum rating" is defined in JIS C 7032 as " a limit value that must not be exceeded even for an instant, and when two or more limits are specified, any two of them must not be reached at the same time.

Taking an input voltage as an example, it is acceptable to apply voltage up to its absolute rated value, but apart from whether the device will operate normally, the resulting behavior will have nothing to do with tolerances, actual performance, or the like. And, exceeding the absolute maximum rated value may possibly cause failure, but this does not mean that the device necessarily will fail for that input voltage. In essence, it is a value that should never under any conditions be exceeded.

This “Confirmation of being within absolute maximum ratings” is based on a proper understanding of these matters.

The confirmation uses the overall switching operation waveform of the MOSFET acquired in the previous section, as well as the waveform data of I_D, V_{DS}, V_{GS}, and power during on-off transition. Basically, we verify that the acquired data does not exceed the absolute maximum ratings of this MOSFET. The absolute maximum ratings and checkpoints for the MOSFET (R6020ENZ) are shown in Table 8. Compare with Figures 40 and 41 in the previous section.

<Summary>

- Properly understand the definition and intent of absolute maximum ratings to determine whether they can be used or not.

4.1.3 ③Confirmation of Being within the SOA (Safe Operating Area)

Flowchart ③. The SOA (Safe Operating Area) is information used to check whether a transistor is operating under safe conditions. In essence, in a graph of the relationship between the drain current I_D and the drain-source voltage V_{DSS}, it is the safe area with respect

4. Absolute maximum ratings [T_a=25°C]

Drain-source voltage		V_{DSS}	・・・ 600V	← Make sure that V_{DS} has not exceeded 600V	
Gate-source voltage		V_{GSS}	・・・ ±20V	static	
			±30V	AC(f>1Hz)	
Drain current	Continuous	I_D	・・・ 20A*	← Make sure that I_D has not exceeded 20A	
	Pulsed	I_{DP}	・・・ ±60A	Pw≦10μs, Duty cycle≦1% ← Make sure that I_{DP} has not exceeded 60A	
Source current (Body diode)	Continuous	I_S	・・・ 20A*		
	Pulsed	I_{SP}	・・・ 60A	Pw≦10μs, Duty cycle≦1%	
Avalanche current		I_{AS}	・・・ 3.4A	L≒70mH, V_{DD}=50V, R_G=25Ω Starting T_{ch}=25℃ See Fig.3-1 and 3-2	
Avalanche energy		E_{AS}	・・・ 418mJ	L≒70mH, V_{DD}=50V, R_G=25Ω Starting T_{ch}=25℃ See Fig.3-1 and 3-2	
Power dissipation		P_D	・・・ 120W	T_c=25℃ ← Make sure that the power dissipation has not exceeded120W	Calculate and check later
Channel temperature		T_{ch}	・・・ 150℃	← Make sure that the channel temperature has not exceeded 150℃	
Range of storage temperature		T_{stg}	・・・ -55~150℃		

Table 8. MOSFET R6020ENZ: Absolute maximum ratings and checkpoints

to the rated voltage and current and the allowable power dissipation (heat generation), that is, the area in which reliability is not impaired and breakdown will not occur. Figure 42 is an SOA graph of the MOSFET R6020ENZ used in this example, and is excerpted from the datasheet.

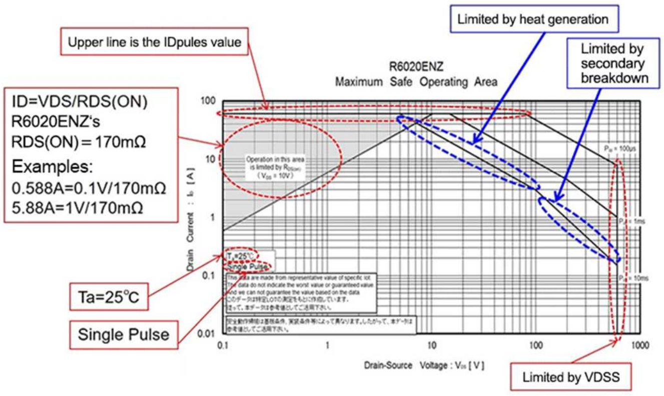


Figure 42. SOA graph of MOSFET R6020ENZ

Using this graph, whether the operating state of a circuit is within the SOA, i.e., safe can be judged. However, there are a number of conditions to be considered when using this data.

- The SOA is only based on a single pulse (in this example, three types of pulse widths are presented). If a transistor is driven by repetitive pulses (that is, continuous switching), it is necessary to check that all the pulses stay within the SOA, and also the average applied power (calculated using Flowchart ⑥) is not greater than the rated power.
- The temperature condition is nearly always Ta＝25℃.
- These are essentially reference values, and not guaranteed values.
- Results will differ depending on transistor mounting conditions and the like.

With such restrictions, it would seem that it would not be consistent under actual usage conditions. However, for example, the thermal resistance of an individual package is a fixed value, but the thermal resistance in a mounted state depends on the mounting conditions. Also, the single-pulse condition would be a very vague condition, even if it is 5 times or 100 times. Therefore, the conditions are simple to some extent and can be clarified. In actuality, the judgment is made by averaging, conversion, and derating, which will be explained later.

4.1.3.1 About SOA Destruction

Conditions outside the SOA during operation can lead to destruction. Using normal operating conditions that are outside of the SOA is out of the question, but overvoltages and short-circuits, which do not occur during normal operation, can cause overcurrent to flow when the applied voltage is maintained, and this can cause momentary local heat generation that results in destruction.

Moreover, inappropriate thermal design or a higher operating frequency can lead to inadequate heat dissipation of a transistor chip, so that continuous heat generation occurs, and if the allowed temperature of the channel is exceeded, thermal runaway can result in destruction (Figure 43).

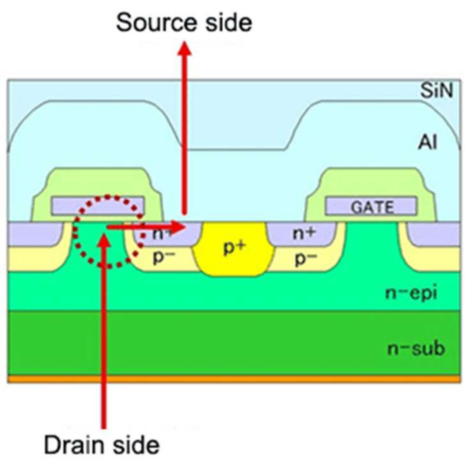


Figure 43. Destruction mode due to thermal runaway of MOSFET channel

<Summary>

- The SOA (Safe Operating Area) is information used to confirm that a transistor is operating under safe conditions.
- In essence, the SOA indicates in a graph the safe area of operation with respect to the rated voltage and current and the allowable power dissipation (heat generation).
- The SOA conditions should be checked carefully, and differences with actual usage conditions should be considered, prior to device use.

4.1.4 ④ Confirmation of Being within the Derated SOA at the Operating Ambient Temperature

Flowchart ④. In the previous section, the concept of SOA was explained, together with the use of graphs and the like, and points to be considered were also discussed. This time, in connection with one of those points for consideration, that the data of a SOA graph is for an ambient temperature Ta of 25℃, we calculate the SOA at the temperature of actual use, and confirm the transistors are used under appropriate conditions.

4.1.4.1 What is SOA Derating?

Derating refers to operating components below their rated values in order to improve reliability. This is also true for SOA. In most cases, the ambient temperature at the actual operating location is higher than 25℃, and the SOA should be the area where the current and voltage are further reduced from the given SOA under 25℃ conditions. Figure 44 shows images of SOA derating for MOSFETs and bipolar transistors.

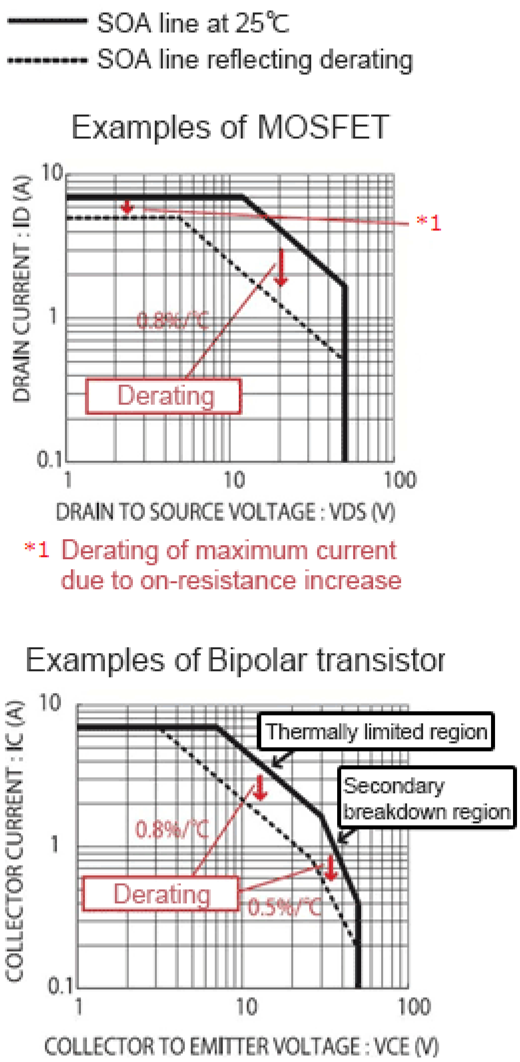


Figure 44. Derating image of SOA

SOA derating is basically considered in terms of allowable power, or heat, and finally the absolute maximum rating of Tj. As a general example, the graph in Figure 44 shows that for MOSFETs, the derating is 0.8%/℃ in the thermal limit region with a maximum current derating due to increased on-resistance. Also, for bipolar transistors, this is an example of derating of 0.8%/℃ in the thermal limit region and 0.5%/℃ in the secondary breakdown region.

As a specific example, Figure 45 shows the power dissipation derating graph for the MOSFET R6020ENZ used as an example and the SOA graph with the derating based on it written in. Each graph is taken from the datasheet.

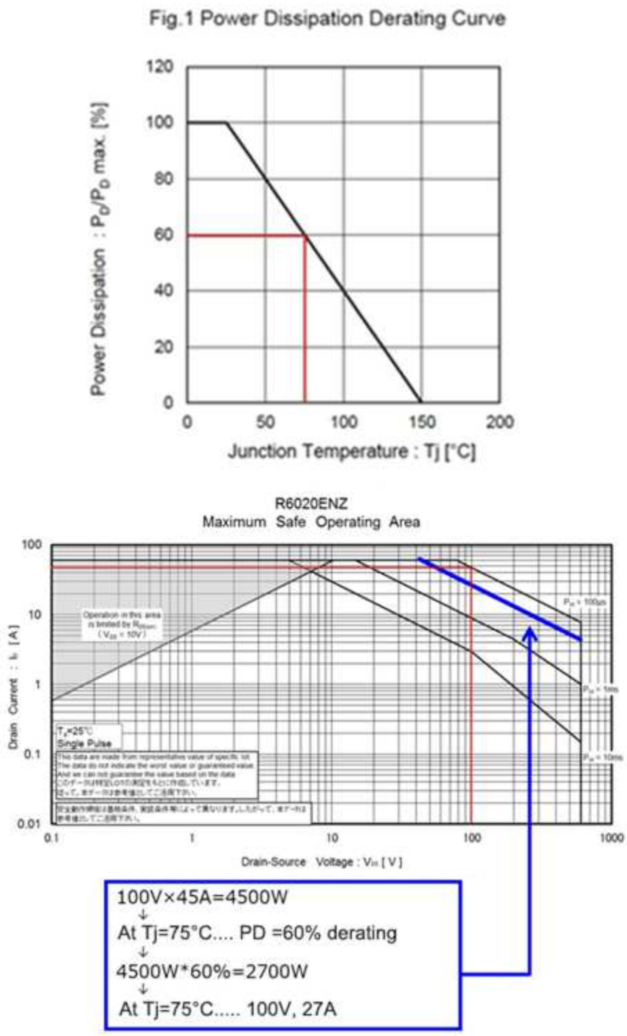


Figure 45. Power dissipation derating graph for R6020ENZ and SOA with derating written in

As an example, assume that the Tj of the actual transistor is 75℃. Previously we had been discussing the ambient temperature, that is, Ta, but when considering allowable losses in a transistor, ultimately it is necessary to consider Tj.

Let us take a moment to examine the relationship with temperature. Among matters requiring consideration that were discussed in chapter 4.1.3 was the condition that "data must be for single pulses". In cases of continuous operation use, conditions are somewhat troublesome, but by stipulating the condition of single pulses, it is possible to set Ta ≈ Tj. This is a general interpretation, based on the consideration that for a pulse of short duration, there is almost no rise in the transistor chip temperature. In parameter tests for semiconductor devices, also including transistors, when heat generation is also considered, in many cases pulse tests assumed that Ta ≈ Tj. Hence it is assumed that the data for this SOA is Ta ≈ Tj ≈ 25℃.

However, for transistors it is necessary to determine Tj from the power dissipation and the package thermal resistance R_{thJA} (θja) and Ta, or else R_{thJC} (θjc) and Tc. These thermal calculations are of a general nature, and so are here omitted.

From the derating curve for the power dissipation in Figure 45, we see at Tj = 75℃ relative to Tj = 25℃, derating to 60% is necessary. One need only use this rate to perform derating as in the calculation example. If necessary, by computing the derating rate, (100%-60%) / |25℃-75℃| = 0.8%/℃, calculations can be performed immediately, and the SOA derating curve may be drawn as indicated by the blue line in the SOA graph.

Finally, the actual conditions of use of the transistor must be confirmed to be within the derated SOA, and whether the usage conditions are appropriate must be judged.

<Summary>

- An SOA graph is for data at Ta=25, and so the SOA must be derated according to the temperature at which the transistor will actually be used.
- As the derating rate, the derating rate for allowable power dissipation is used.

4.1.5 ⑥Confirmation that Average Power Consumption is within the Rated Power

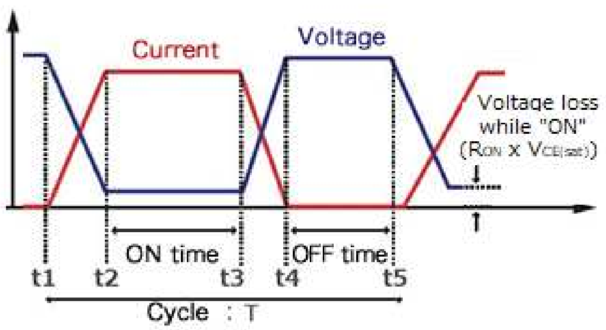
Flowchart ⑥. Although ⑤ is skipped, ⑤ is a confirmation of single pulse or continuous pulse, and since this example is a switching operation, simply select the flow for continuous pulse and immediately start considering ⑥.

We determine the average power consumption for a case in which the operation is switching operation (continuous pulse), and check whether the operation is within the rated range. This is necessary because, as explained in Flowchart ③, the SOA conditions are for "single pulse" operation, and so cannot be applied directly to a case of continuous pulse operation, such as switching operation. Hence as an additional check, we confirm that power consumption is within the rated range. In the case of switching operation, there is repetition of the transistor-on state in which current flows and power is consumed and the off-state in which there is no power consumption, and so it is necessary to determine the average power consumption.

4.1.5.1 Method of Determining Average Power Consumption during Switching Operation

The average power consumption can be calculated by totaling the integrated values of the power, which is the product of the voltage and the current, for the interval of the transition from the transistor off-state to the on-state (t1-t2), the transistor on-time (t2-t3), the transition from on to off (t3-t4), and the off time (t4-t5), and dividing the result by the cycle (T). Please refer to Figure 46 diagram and equation.

$$P = \frac{\int_0^t I(t)V(t)dt}{T} \ (W)$$



Case of switching operation as in the diagram above:

$$P = \frac{\int_{t1}^{t2} IVdt + \int_{t2}^{t3} IVdt + \int_{t3}^{t4} IVdt + \int_{t4}^{t5} IVdt}{T} \ (W)$$

Figure 46. Method of determining average power consumption during switching operation

The switching voltage/current waveform data acquired in Flowchart ① "Measurement of Actual Current and Voltage Waveforms" is used here (Figure 47).

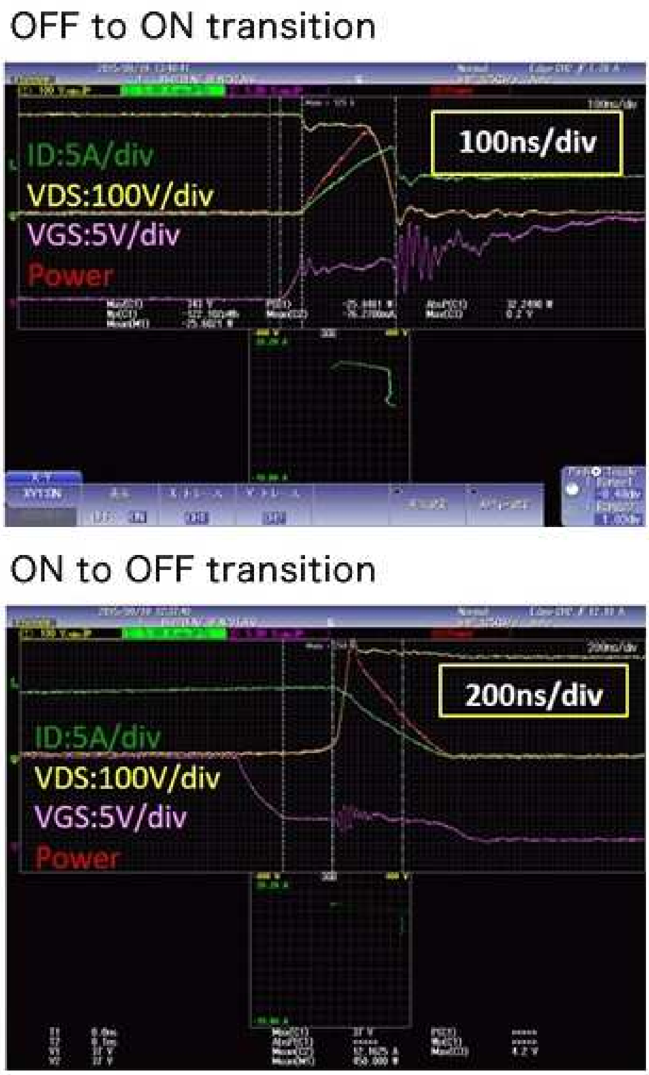


Figure 47. Waveform data acquired in ① Measurement of Actual Current and Voltage Waveforms

Based on these waveforms, we perform calculations in accordance with the integral equation in Figure 48. This is for the example of a MOSFET, but if the device were a bipolar transistor, the same calculations could be performed, using the collector current I_C and the collector-emitter voltage V_{CE}.

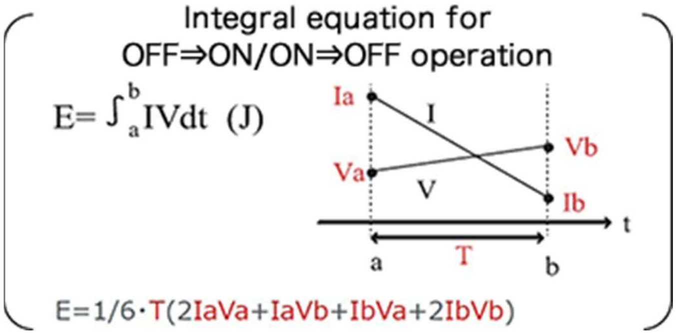


Figure 48. Integral formula for finding average power consumption from waveforms

If the average power consumption obtained through these calculations is within the maximum rated value for allowable power dissipation given on the data sheet, then we judge that losses do not pose a problem. Of course, it is necessary to determine whether the result is within the rated value for the actual operating temperature.

The absolute maximum rating of Power Dissipation PD for MOSFET R6020ENZ is 120W as shown in Table 8 in Flowchart ②, " Confirmation of Being within Absolute Maximum Ratings," but since the condition is Tc = 25℃, this value simply cannot be applied under other conditions. For this reason, the derating graph of allowable power dissipation PD in Figure 45, which is also used in Flowchart ④, "Confirmation of Being within the Derated SOA at the Operating Ambient Temperature," is used. Basically, this graph is used to determine if the product can be used in the form of converting power consumption to Tj.

In any case, since Tj is to be determined, it may be more straightforward to confirm that Tj does not exceed the absolute maximum rating of 150℃ for Tj under the operating temperature conditions. Since the power consumption is known, the calculation is easy once the thermal resistance of the package is known: R_{thJC} (θjc) for MOSFET R6020ENZ is 1.04℃/W Max and R_{thJA} (θja) is 40℃/W Max.

The calculations are basically calculations to determine Tj, where Tj = power consumption × thermal resistance + operating temperature (maximum). There is no problem with using either Ta or Tc as the operating temperature, according to convenience.

There are a number of methods for confirming that the power consumption is within the rated value, but they are in the end equivalent to determining whether Tj is within the absolute maximum rated value. In the next section, the calculation of Tj is discussed in more detail below.

<Summary>

- In the case of continuous-pulse (switching) operation, the average power consumption is determined, and it is confirmed that allowable power dissipation is within the rated value.
- A judgment will in the end depend on whether Tj exceeds the absolute maximum rating.

4.1.6 ⑦Confirmation of the Chip Temperature

Although not included in the flowchart, the last step ⑦ is to confirm the chip temperature.

As noted at the end of the previous section, whether the power consumption is within the rating is ultimately a matter of whether Tj is within the absolute maximum rating. Therefore, it is necessary to know how to determine Tj = chip temperature (junction temperature).

In actuality, if the power consumption is within the rated value, then the chip temperature is below the absolute maximum rated value. This is because, as mentioned above, the power rating is calculated on the basis of the chip temperature.

Hence whether usage conditions are appropriate can be determined by the confirmations made up to Flowchart ⑥. However, we have added the further step of confirming the chip temperature because, in addition to further heightening certainty, it serves as a firmer foundation when considering cases in which the rated values are exceeded.

One factor determining the chip temperature is the thermal resistance. Although this is explained below in terms of an equation, the relation is simple: when the thermal resistance is lowered, the chip temperature falls. In general, heat dissipation design is executed such that, within the constraints of the conditions of use, the required power is obtained while keeping the chip temperature Tj within the maximum rated value. In general, the power that can be handled given the thermal resistance of a lone transistor is far lower than the performance of which the transistor is capable. In particular, a power transistor is generally used together with a heat sink. In the heat dissipation design for such a transistor, the chip temperature Tj is the most reliable parameter.

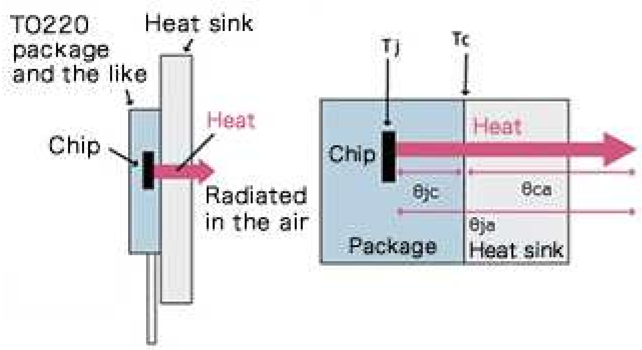
4.1.6.1 Calculation of the Chip Temperature

In recent years many transistors, such as the MOSFET used in examples, are resin-sealed. Consequently, it is of course not possible to measure the chip temperature directly. Hence Tj is essentially determined through calculations. The equation used is as follows. Where Tj: chip (junction) temperature, Ta: ambient temperature (℃), θja (T_{thJA}): thermal resistance, junction to ambient (℃), P: power consumption (W).

$$\mathrm{Tj} = \mathrm{Ta} + \theta\mathrm{ja} \times \mathrm{P}$$

As the equation indicates, the chip temperature Tj is obtained by adding the ambient temperature Ta to the product of the thermal resistance and the power, that is, the chip heat generation. This is the most basic equation for Tj.

Figure 49 indicates the relationship between temperature and thermal resistance for different parts in mounting. The diagram includes a heat-dissipating plate (heat sink), but the case (package) temperature Tc is the temperature of the transistor package surface.



$$\boldsymbol{\theta\mathrm{ja} = \theta\mathrm{jc} + \theta\mathrm{ca}}$$

θja: Thermal resistance, junction - ambient (℃/W)

θjc: Thermal resistance, junction - case (℃/W)

θca: Thermal resistance, case - ambient (℃/W)

Figure 49. Relationship between temperature and thermal resistance of each part in mounting

The thermal resistance θca when there is no heat sink is simply the result of subtracting θjc from the given value of θja. When a heat sink is present, the thermal resistance of the heat sink corresponds to θca, and conversely, θja is not simply the value given, but is the result of adding θjc and θca. Nearly all power transistors use heat sinks, and so in general θja and θjc are given the data sheet.

In general, θjc is quite lower compared with θja, and so θja can be reduced considerably depending on the thermal resistance of the heat sink.

θja of the MOSFET R6020ENZ itself is 40℃/W, but if for example a heat sink with a thermal resistance of 10℃/W is used, then θja can be reduced to 11.04℃/W *. Moreover, when the heat sink is assumed to be infinite in expanse, it can be assumed that θja = θjc. (* The result of a simple calculation for purposes of illustration. An actual θca must be determined considering the thermal resistance at the junction between the package surface and the heat sink. Moreover, the thermal resistance of a heat sink is selected by calculating the value necessary for Tj to remain within the rated value.)

Once the numerical values needed for the calculation are obtained, the calculation itself is extremely simple. The power consumption is based on the average power consumption discussed aforesaid, taking the worst-case conditions into consideration. Ta should be measured directly. However, measurement of Ta is unexpectedly difficult. In measurements a thermocouple may be used to measure the ambient temperature, but the temperatures near a heating element and far away from the heating element differ considerably. Moreover, in actual equipment the module with a power transistor is installed within a case, where air is flowing due to an air-cooling fan, and so it is in fact quite difficult in practice to decide on the conditions under which to measure the ambient temperature Ta. As a result, often Tc is used to determine Tj.

4.1.6.2 Method for Determining Tj from Tc

In recent years, with the widespread use of thermography, radiation thermometers and the like, it has become relatively easy to measure the temperature of a package surface. As the Tc used in calculations, the highest temperature on the package surface is used, and so thermography is a very convenient method for determining at a glance the highest temperature on the package surface.

This method is effective for surface-mounted packages, but for through-hole components such as TO-220 type, a somewhat different approach is needed. In the case of a TO-220 type power transistor, in actual use the device is nearly always paired with a heat sink, and the heat sink is installed on the rear surface of the package. Hence the case (package) temperature (meaning the “c” in θjc, rather than the bottom or top) in a case such as that shown above becomes the temperature at the junction with the heat sink. Because the package is accompanied by the heat sink, it should be assumed that thermography cannot be used to measure the case temperature (at the junction with the heat sink). One alternative is to use thermocouples. A hole can be drilled in the heat sink and the temperature can be measured by placing a thermocouple directly on the backside of the package.

Or, one could calculate backward from the temperature of the heat sink. In this case, rather than the θca of the heat sink, the thermal resistance of the heat sink itself would be necessary. Or, it might be simpler to return to basics and perform calculations starting from Ta. It is important that Tj be known accurately, but in any case a margin must be provided, and so some thought should be given to how much precision is needed.

Below is the formula for obtaining Tj from Tc. It is based on the thermal resistance relationship shown in Figure 49. Where Tc: maximum case (package) surface temperature (℃), θjc: thermal resistance, junction to case surface (℃/W), P: power consumption (W).

$$\mathrm{Tj} = \mathrm{Tc} + \theta\mathrm{jc} \times \mathrm{P}$$

Thus if Tj does not exceed the maximum rating of 150℃, it can be concluded that the selected transistor is suitable. As noted at the beginning of this section, if Tj exceeds 150 ℃ , it can be addressed by reducing thermal resistance, such as by changing the heat sink based on the above calculations.

<Summary>

- It should be confirmed that Tj ultimately does not exceed the absolute maximum rating.
- Tj is the sum of Ta or Tc and the heat generation (the product of the thermal resistance and the power consumption).

Although it is time-consuming work to perform each of the check items in the "Confirming Transistor Suitability in Actual Operation” section, please understand that it is an unavoidable process in circuit design.

Conclusion

As Si-based power devices, we picked up rectifier diodes, Schottky barrier diodes (SBDs), fast recovery diodes (FRDs), MOSFETs, and super junction MOSFETs (SJ MOSFETs) and explained their basic characteristics. It also shows the method and procedure to determine if the selected transistor is appropriate under actual operating conditions. In addition, a list of terms related to MOSFET specifications is included on the next page as an appendix.

In particular, designs that use power devices require an understanding of the direct characteristics of discrete components. In addition, since power device evaluation deals with high power, it requires a certain amount of instrumentation, equipment, and safety assurance. In recent years, with the development of simulation, it has become relatively easy to perform thermal simulation in addition to electrical simulation. Of course, verification with actual equipment is essential, but the use of simulation in the design process is a highly effective approach that should be considered. ROHM offers a web-based simulation tool "ROHM SOLUTION SIMULATOR" that can be used free of charge.

https://www.rohm.com/solution-simulator

<Appendix> Terms Related to MOSFET Specifications

The specifications (data sheets) of electronic components, including MOSFETs, contain electrical specifications, parameter names, guaranteed values, etc. MOSFETs also have many parameters, which are listed here.

Note that parameter names, terms, and symbols may differ slightly from manufacturer to manufacturer. There are also some differences in definitions related to condition setting. These should be addressed by checking the measurement conditions and other information provided in the specifications to understand the specifics.

● Absolute Maximum Ratings

Parameter	Symbol	Definition and Description
Drain-source voltage	V_{DSS}	The maximum value of the voltage that can be applied across the drain and source with the gate and source put into a short-circuit state.
Drain current (direct)	I_D	The maximum value of the direct current that can flow continuously in the channel formed between the drain and source under specified conditions.
Drain current (pulse)	I_{DP}	The maximum value of the current that can flow in pulses in the channel formed between the drain and source, for a pulse width and duty ratio specified by the safe operating area.
Gate-source voltage	V_{GS}	The maximum value of the voltage that can be applied across the gate and source.
Avalanche current (single pulse)	I_{AS}	The maximum value of the drain current that is allowed during avalanche breakdown.
Avalanche energy (single pulse)	E_{AS}	The maximum value of the energy that is allowed during avalanche breakdown.
Power dissipation	P_D	The maximum value of the power dissipation allowed in the MOSFET under specified conditions.
Junction temperature	T_j	The maximum value of the junction temperature that is allowed during component operation.
Storage temperature	T_{stg}	The temperature range within which the device can be stored or transported without applying an electrical load to the component.

● Thermal Resistance

Parameter	Symbol	Definition and Description
Thermal resistance (junction–case)	R_{thJC}	The thermal resistance value from the device junction to the rear surface of the case.
Thermal resistance (junction–ambient)	R_{thJA}	The thermal resistance value from the device junction to the ambient environment.
Soldering temperature (wave soldering)	T_{sold}	The maximum value of the solder melting temperature when mounting a MOSFET.

● **Electrical Characteristics**

Parameter	Symbol	Definition and Description
Drain–source breakdown voltage	$V_{(BR)DSS}$	In the state in which the gate and source are short-circuited, the voltage at which breakdown of the parasitic diode occurs and a current begins to flow between the drain and source.
Drain–source breakdown voltage temperature coefficient	$\Delta V_{(BR)DSS}/\Delta T_j$	The temperature coefficient of the drain-source breakdown voltage.
Zero gate voltage drain current	I_{DSS}	The leakage current that flows between the drain and source when the gate and source are short-circuited under specified conditions.
Gate leakage current	I_{GSS}	The leakage current that flows between the gate and source when the drain and source are short-circuited under specified conditions.
Gate threshold voltage	$V_{GS(th)}$	The gate-source voltage at which a drain current begins to flow in the MOSFET.
Gate threshold voltage temperature coefficient	$\Delta V_{GS(th)}/\Delta T_j$	The temperature characteristic of the gate threshold voltage.
Drain–source ON resistance	$R_{DS(on)}$	The value of the resistance between the drain and source when the MOSFET is turned on.
Gate resistance	R_G	The internal gate resistance value of the MOSFET.
Forward transfer admittance	$\|Y_{fs}\|$	The rate of change in the drain current with a 1 V change in the gate-source voltage.
Input capacitance	C_{iss}	The value of the parasitic capacitance between the gate and source, measured in the state in which the drain and source are short-circuited for alternating currents.
Output capacitance	C_{oss}	The value of the parasitic capacitance between the drain and source, measured in the state in which the gate and source are short-circuited for alternating currents.
Feedback capacitance	C_{rss}	The value of the parasitic capacitance between the drain and gate, measured with the source pin grounded.
Effective capacitance (referred to energy)	$C_{o(er)}$	The fixed capacitance value equivalent to the C_{oss} value resulting in the energy accumulated from when the drain-source voltage V_{DS} is 0 V until V_{DS} rises to 80% of the absolute maximum rating of the drain-source voltage.
Effective capacitance (referred to time)	$C_{o(tr)}$	The fixed capacitance value equivalent to the C_{oss} value resulting in the charging time from when the drain-source voltage V_{DS} is 0 V until V_{DS} rises to 80% of the absolute maximum rating of the drain-source voltage.
Turn-on delay time	$t_{d(on)}$	The time from when the gate-source voltage has risen to 10% of a voltage setting until the drain-source voltage has fallen to 90% of a set voltage.
Rise time	t_r	The time required for the drain-source voltage to fall from 90% to 10% of a voltage setting.
Turn-off delay time	$t_{d(off)}$	The time from when the gate-source voltage has fallen to 90% of a voltage setting until the drain-source voltage has risen to 10% of a set voltage.
Fall time	t_f	The time required for the drain-source voltage to rise from 10% to 90% of a voltage setting.

● **Gate Charge Characteristics**

Parameter	Symbol	Definition and Description
Total gate charge	Q_g	The amount of gate charge necessary to raise the gate voltage of the MOSFET from 0 V to a specified voltage.
Gate–source charge	Q_{gs}	The amount of charge accumulated on the capacitance between the gate and source that is necessary to raise the gate voltage of the MOSFET from 0 V to the gate plateau voltage.
Gate–drain charge	Q_{gd}	The amount of charge accumulated on the capacitance between the gate and drain that is necessary for the MOSFET drain-source voltage V_{DS} to fall from the power supply voltage to the ON voltage.
Gate plateau voltage	$V_{(plateau)}$	The value of the gate voltage at which mirror capacitance charge/discharge begins during switching.

● **Internal Diode Electrical Characteristics**

Parameter	Symbol	Definition and Description
Source current (direct)	I_s	The maximum value of the direct current that can flow through the internal diode continuously under specified conditions.
Source current (pulse)	I_{sp}	The maximum value of the current that can flow in pulses in the internal diode.
Forward voltage	V_{SD}	The drop in voltage when a forward current has flowed in the internal diode.
Reverse recovery time	t_{rr}	The time required for the reverse recovery current in the internal diode to vanish under specified measurement conditions.
Reverse recovery charge	Q_{rr}	The amount of charge required for the reverse recovery current in the internal diode to vanish under specified measurement conditions.
Peak reverse recovery current	I_{rrm}	The value of the peak current during reverse recovery operation of the internal diode under specified measurement conditions.

Revision History

Date	Version	Details
2023.8.21	001	Initial version in English. Translation of Japanese version 002.

Notes

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